SkelGym: Cross-Paradigm Kinematic Fusion and Geometry-Preserving Skeletal Augmentation for 22-Class Gym Exercise Classification

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Abstract

Automated gym exercise recognition is critical for developing intelligent virtual personal trainer systems capable of monitoring workout execution, tracking training volume, and mitigating musculoskeletal injury risks. However, traditional RGB video approaches incur prohibitive computational costs, suffer from environmental background overfitting, and raise severe privacy concerns when streaming from home environments. We evaluate on a curated corpus of 1,024 in-the-wild videos spanning 22 fine-grained resistance exercises under strict video-level partitioning with no source-video overlap across train, validation, and test sets. We present **SkelGym**, a lightweight, privacy-aware intelligent exercise recognition framework operating exclusively on 3D skeletal coordinates extracted via monocular pose estimation. SkelGym introduces four primary technical contributions: (1) a curated, multi-source 3D skeletal benchmark spanning 1,024 videos across 22 resistance exercises under strict video-level partitioning with zero split leakage; (2) a 63-dimensional *Biomechanical Compound Representation* (Biomechanical Mix v2) fusing pelvic-centered metric World 3D coordinates (39-d) with 24 domain-selected kinematic angles (14 segment elevation and 10 joint articulation angles) via dual-branch projection; (3) *SkelGym-Aug*, an empirically selected, task-oriented augmentation configuration restricted to distance-preserving Euclidean transformations (sagittal bilateral reflection and gravitational yaw rotation) se-

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lected strictly on validation-window evidence; and (4) a cross-paradigm ensemble coupling a self-attention Transformer with a four-stream Adaptive Graph Convolutional Network via zero-parameter uniform average soft voting (SkelGym-Full) and validation-calibrated accuracy-weighted soft voting (SkelGym-Lite), achieving real-time low-latency edge classification. Evaluated across three independent random seeds under fixed splits ($N = 2,743$ test windows from 233 held-out videos), SkelGym achieves **76.19% ± 0.16% window-level accuracy** (Window Macro F1: **0.7537 ± 0.0014**) and **85.27% ± 0.41% video-level consensus accuracy** (Video Macro F1: **0.8386 ± 0.0075**, correctly classifying 200 of 233 held-out videos, achieving 85.84% accuracy on the canonical test run). All pairwise architectural transitions are verified via paired statistical hypothesis testing (window-level McNemar $p \leq 5.88 \times 10^{-5}$ across all transitions; video-level posterior confidence differences confirmed across sequence-to-ensemble transitions). With an ultra-low classifier inference latency of merely **0.49–3.96 ms** on host CPU (0.59–4.65 ms on CUDA, where individual backbones execute in 0.49–1.03 ms and the full five-stream ensemble executes in 3.96 ms at 253 FPS) per 32-frame window, SkelGym enables real-time, privacy-aware exercise recognition suitable for smart mirrors, mobile devices, and embedded gym displays. All source code, extracted skeletal representations, metadata, and pretrained checkpoints used for reproducibility are publicly available at `https://huggingface.co/Cuong2004/gym-exercise-classification` co the phai chuyen link sang github.

Keywords: Gym exercise classification, Skeleton-based action recognition, Adaptive Graph Convolutional Networks, Skeletal Transformers, Biomechanical features, Real-time inference, Edge AI

1. Introduction

Regular physical resistance training and structured exercise play an essential role in improving cardiovascular health, metabolic longevity, muscular hypertrophy, and psychological well-being [39]. With modern demanding lifestyles and remote working environments, a rapidly growing population conducts workouts in home settings rather than commercial fitness centers. However, exercising without the supervision of certified strength and conditioning coaches frequently leads to improper kinetic execution, sub-optimal muscular recruitment, and elevated risks of acute or repetitive strain in-

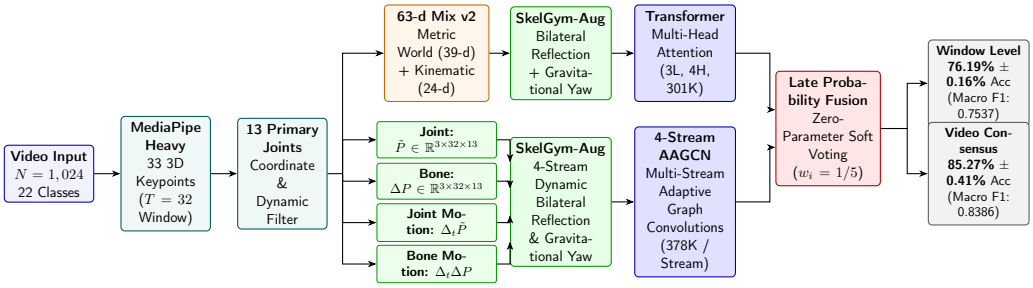


Figure 1: The proposed SkelGym architecture: an end-to-end cross-paradigm framework coupling self-attention sequence modeling with multi-stream adaptive graph convolutional networks via zero-parameter uniform average soft voting.

juries (such as lumbar disc herniation, rotator cuff impingement, and patellar tendinopathy) [34]. Consequently, there is an urgent demand for automated, intelligent virtual personal trainers capable of identifying exercise types in real time, monitoring execution repetitions, and providing instantaneous biomechanical feedback [49].

Early computer vision frameworks for resistance exercise and action recognition were proposed upon appearance-based RGB video models, such as 2D/3D CNNs [1, 2, 3] and spatiotemporal Vision Transformers (e.g., ViViT [4], TimeSformer [5], Video Swin [6]). While dense RGB representations capture visual cues such as gym equipment and barbells [7], they exhibit three critical operational drawbacks in practical deployments. First, dense spatio-temporal convolutions impose heavy computational overhead (often exceeding hundreds of GFLOPs), preventing real-time execution on resource-constrained edge devices or mobile hardware. Second, appearance-based networks are readily dominated by background visual cues, flooring textures, and athletic apparel, causing severe performance degradation when evaluated in novel domestic workout environments. Third, cloud-centric video architectures require continuous transmission of high-definition RGB video streams from private living spaces to remote servers, introducing network bandwidth bottlenecks and acute user privacy risks.

To address these challenges, skeleton-based human action recognition offers an elegant solution by abstracting human bodily movement into sparse, lightweight 3D keypoint coordinates. With real-time pose estimation frameworks such as Google MediaPipe [8, 9], OpenPose [10], and AlphaPose [11], 3D skeletal tracking can be executed on edge devices and commodity hardware at sub-millisecond latencies. However, translating skeletal tracking into

accurate, multi-class gym exercise recognition introduces three fundamental scientific challenges:

The first challenge is balancing biomechanical redundancy against kinematic expressiveness. Distinguishing fine-grained exercise variants (such as flat, incline, and decline bench presses, or differentiating conventional deadlifts from Romanian deadlifts) requires capturing subtle joint-angle dynamics and limb orientations. However, exhaustive geometric angle formulations—such as computing all three-joint triplets—incur a combinatorial feature explosion ($\binom{13}{3} = 286$), introducing high dimensional multicollinearity, gradient singularities at joint lockouts, and severe overfitting risks.

The second challenge stems from physiologically invalid data augmentation. While data augmentation is standard in general action recognition, applying unconstrained spatial or temporal transformations naively to resistance training distorts invariant anatomical bone lengths, violates gravitational verticality, or corrupts movement execution cadence and velocity profiles that define distinct exercise classes.

Finally, the third challenge lies in achieving true cross-paradigm model complementarity while strictly preventing data contamination. Temporal sequence architectures, such as LSTMs [12, 13] and Transformers [14], excel at modeling long-range phase transitions. Whereas spatio-temporal graph convolutional networks, such as ST-GCN [16], 2S-AGCN [17], and CTR-GCN [20], explicitly exploit physical musculoskeletal topology. Prior works often combine these paradigms naively, rely on heavy pre-trained backbones unsuitable for edge hardware, or tune ensemble fusion weights directly on test partitions, inducing optimistic reporting bias.

To systematically resolve these challenges, SkelGym (illustrated in Fig. 1), a scientifically grounded cross-paradigm skeletal classification framework, is proposed. SkelGym transforms raw monocular video into actionable exercise labels through a fully automated pipeline: *Video* $\rightarrow$ *MediaPipe 3D Skeleton* $\rightarrow$ *63-d Biomechanical Representation* $\rightarrow$ *SkelGym-Aug* $\rightarrow$ *Cross-Paradigm Classifiers* $\rightarrow$ *Late Probability Fusion* $\rightarrow$ *Video Consensus* $\rightarrow$ *Edge Deployment*. Specifically, the primary technical contributions of this work are as follows:

- **C1: Multi-Source 3D Skeletal Landmark Benchmark with Leakage-Free Partitioning.** We curate and release a fine-grained 3D skeletal dataset derived from 1,024 multi-camera in-the-wild videos spanning 22 resistance training exercises across three provenance streams:

public repositories, licensed stock media, and 128 ethically collected author-recorded gym recordings. Pre-extracted into 13 load-bearing 3D landmark trajectories (1,108 action segments) in JSON, NPY, and HDF5 formats under CC BY 4.0 and MIT licenses, our benchmark establishes a strict video-level partitioning protocol (6:2:2 train/val/test split) ensuring zero cross-split source video overlap, completely eliminating temporal window-level data leakage across partitions.

- **C2: Biomechanical Compound Representation (63-d Mix v2).** We design an upgraded kinematic descriptor fusing pelvic-centered metric World 3D coordinates (39-d) with 24 domain-selected kinematic angles. Spatial coordinates are expressed in estimated physical metric dimensions (meters) centered at the pelvic mid-hip origin anchor. The angular branch distills 24 anatomically decisive angles (14 limb segment orientation angles relative to gravity and 10 3D joint articulation triplets), resolving collinearity and gradient singularities while providing intrinsic invariance to horizontal camera yaw rotation.

- **C3: Empirically Selected Task-Oriented Augmentation Configuration (SkelGym-Aug).** Rather than claiming novel mathematical transformation operators, we evaluate a candidate pool of established skeletal augmentations categorized into geometric, temporal, and noise-based domains. Through systematic Leave-One-Out (LOO) and isolated ablations evaluated strictly on validation-window evidence, we demonstrate that unconstrained scaling and jitter distort physical metric proportions and velocity profiles, while temporal time warping disrupts exercise cadence. Consequently, we isolate an optimal, task-oriented 2-operator configuration—Bilateral Sagittal Reflection and Gravitational Yaw Rotation ($\pm 15^{\circ}$)—that strictly preserves physical Euclidean distances and gravitational verticality, achieving peak representation learning (Validation Window Accuracy: 81.20% $\pm$ 0.72%, Macro F1: 0.8119 $\pm$ 0.0072) with zero test-time data corruption.

- **C4: Cross-Paradigm Fusion, Rigorous Benchmarking, and Real-Time Edge Deployment.** We engineer custom lightweight backbones ($\approx$ 301K–378K parameters each, trained entirely from scratch) spanning self-attention sequence Transformers and four-stream Adaptive Graph Convolutional Networks (AAGCN), unified via transpar-

ent late probability fusion: zero-parameter uniform average soft voting for the full cross-paradigm ensemble (SkelGym-Full) and validation-calibrated accuracy-weighted soft voting for the two-stream edge variant (SkelGym-Lite). Evaluated across three independent random seeds $(42, 123, 3407)$ and paired statistical tests (window-level McNemar $p \leq 5.88 \times 10^{-5}$ across all transitions; video-level posterior confidence differences confirmed across sequence-to-ensemble transitions), SkelGym attains **76.19% ± 0.16% window accuracy** (Window Macro F1: **0.7537 ± 0.0014**) and **85.27% ± 0.41% video consensus accuracy** (Video Macro F1: **0.8386 ± 0.0075**, correctly classifying 200 of 233 held-out test videos). Delivering an ultra-low single-window classifier latency of merely **0.49–3.96 ms** on host CPU (0.59–4.65 ms on CUDA, where individual backbones execute in 0.49–1.03 ms and the full 5-stream ensemble executes in 3.96 ms at 253 FPS), SkelGym demonstrates practical real-time feasibility for privacy-aware edge deployment without persistent video streaming.

2. Related Work

2.1. Skeletal Action Recognition

Early landmark-based action recognition utilized Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks [12, 13, 15] to model skeletal temporal dynamics. However, pure recurrent models treat skeletons as unorganized coordinate vectors, ignoring the inherent topological structure of the human musculoskeletal system. To overcome this, Yan et al. [16] pioneered Spatial Temporal Graph Convolutional Networks (ST-GCN), constructing a spatio-temporal graph where human joints form vertices and physical bones form fixed spatial edges. Shi et al. [17] introduced Two-Stream Adaptive Graph Convolutional Networks (2S-AGCN), parameterizing the adjacency matrix with learnable and data-driven components. Subsequent advances, including multi-stream AAGCN [18], CTR-GCN [20], InfoGCN [47], HD-GCN [48], and SkateFormer [21], progressively integrated channel attention, information-theoretic representation learning, hierarchical graph decomposition, and partition-specific transformer attention. In parallel, Duan et al. [56] introduced PoseConv3D, mapping 3D skeleton keypoints to spatiotemporal heatmap volumes to exploit 3D convolutional architectures, while Yin et al. [50] designed an efficient multi-stream depthwise separable convolutional network for skeleton action recognition. However,

these modern architectures were predominantly tuned around large-scale benchmarks like NTU RGB+D (60/120 classes) with parameter footprints of 1.3M–5.2M and high GFLOP demands. In resistance exercise tracking, peripheral joints suffer severe occlusion by barbells and machine frames. SkelGym prunes the skeleton to $V = 13$ load-bearing joints and designs custom lightweight backbones ($\approx$ 350K parameters per stream) trained from scratch, delivering real-time edge execution.

2.2. Kinematic Sequence Modeling and Skeletal Transformers

Following the success of self-attention in natural language processing and computer vision [14, 4, 5, 6], transformers have been adapted for skeletal motion [19]. However, despite global receptive fields, transformers are data-hungry and prone to overfitting on small skeletal datasets. We demonstrate that when fed with our 63-d Biomechanical Mix v2 Representation and regularized by SkelGym-Aug, a compact 3-layer Transformer (301K parameters) achieves competitive performance, functioning as an ideal complement to graph convolutions.

2.3. Skeletal Data Augmentation in Action Recognition

Data augmentation is essential for mitigating overfitting in deep skeleton-based action recognition, particularly under constrained training regimes [16, 18]. Existing augmentation strategies in the literature typically fall into three broad categories: spatial-geometric transformations (random 3D rotations, sagittal flipping, coordinate scaling) [16, 18], temporal manipulations (temporal cropping, time warping, speed perturbation) [22], and noise injection (additive Gaussian coordinate jitter) [23]. Although these general transformations enhance model robustness in generic action recognition, they often ignore domain-specific biomechanical constraints: unconstrained 3D rotations disrupt gravitational alignment, non-uniform spatial scaling distorts invariant anatomical bone proportions, and arbitrary temporal manipulation risks corrupting movement execution dynamics and cadence. In this work, we formulate a disciplined candidate pool across these three categories and conduct systematic Leave-One-Out (LOO) and isolated single-component ablations evaluated strictly on validation-window evidence. This analysis reveals that while distance-preserving Euclidean isometries (bilateral sagittal reflection and gravitational yaw rotation) supply essential viewpoint and bilateral invariance, non-rigid operators (scaling, coordinate jitter, and temporal time

warping) alter anatomical proportions and cadence, degrading representation quality. The resulting empirically selected *SkelGym-Aug* configuration (Mirror + Yaw) provides a tailored, biomechanically grounded regularizer for gymnasium motion analysis.

2.4. Fitness and Resistance Exercise Recognition

Recently, automated exercise tracking systems have attracted growing attention in sports science, rehabilitation, and ubiquitous computing. García-de-Villa et al. [49] developed an expert system for simultaneous exercise recognition and evaluation in prescribed fitness routines for virtual coaches. Strömbäck et al. [51] created MM-Fit, a multimodal benchmark unifying 3D pose, multi-view RGB-D video, and wearable inertial sensors across free-weight exercises, demonstrating the utility of multi-sensor tracking. Riccio [34] implemented real-time exercise classification and repetition counting using a BiLSTM over geometric joint angles. Ding et al. [36] developed the RepCount benchmark for repetition counting in diverse fitness environments. Deyzel et al. [46] proposed one-shot skeleton-based recognition for strength and conditioning exercises on the SU-EMD dataset, revealing critical confusion between squats and deadlifts due to kinematic overlap. While general benchmarks (UCF101 [37], Kinetics-400 [38]) feature broad athletic activities, they lack fine-grained biomechanical resolution. Existing fitness datasets are either restricted to trivial 3–5 exercises or employ random temporal window splits that permit data leakage. In contrast, SkelGym benchmarks a demanding 22-class resistance exercise taxonomy under a strictly segregated video-level split protocol, advancing beyond laboratory-controlled settings to in-the-wild multi-camera gym environments.

3. Dataset and Experimental Protocol

3.1. Data Source, Provenance, and Ethical Compliance

The experimental dataset consists of 3D skeletal landmarks extracted from 1,024 clean, un-shuffled video recordings across 22 resistance training exercises: *barbell biceps curl, bench press, chest fly machine, deadlift, decline bench press, hammer curl, hip thrust, incline bench press, lat pulldown, lateral raise, leg extension, leg raises, plank, pull-up, push-up, Romanian deadlift, Russian twist, shoulder press, squat, t-bar row, tricep dips, and tricep pushdown* (Fig. 2). To ensure high diversity across environmental backgrounds, camera perspectives, illumination conditions, and athlete anthropometry, the

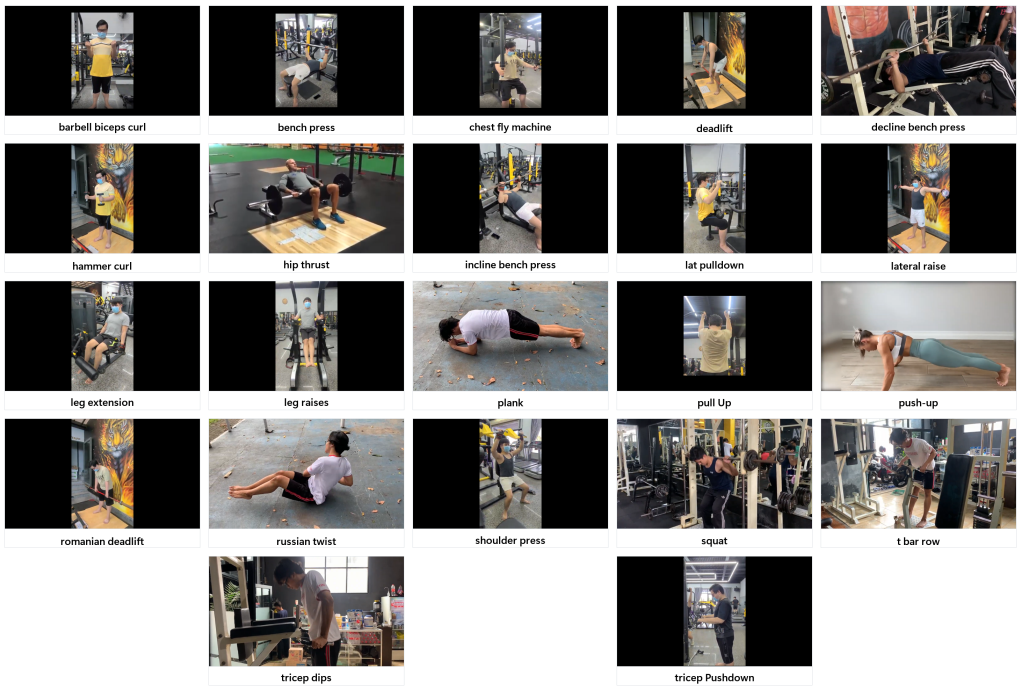


Figure 2: Taxonomy and sample distributions across the 22 resistance exercises in the benchmark dataset.

dataset unifies three distinct provenance streams: (1) public video repositories, comprising 652 videos recovered from the open Workout/Exercise Video Dataset [35]; (2) curated online media, including 103 high-definition clips collected from YouTube fitness channels, 65 clips from Pexels, and 76 clips from Freepik acquired under open research and stock licenses; and (3) author-recorded footage, consisting of 128 supplementary videos captured by the authors in a gymnasium across all 22 exercise classes with two athletic models, employing both static tripod viewpoints and dynamic moving-camera angles to capture multi-perspective kinetic execution.

Ethical Standards, Subject Diversity, and Data Integrity.. The benchmark combines two complementary data streams: 128 ethically collected gymnasium videos recorded under controlled multi-angle setups with written informed consent (Declaration of Helsinki), and 896 web-sourced videos capturing diverse athletes across unconstrained environments. To prevent demographic bias, the dataset explicitly omits participant identity and demo-

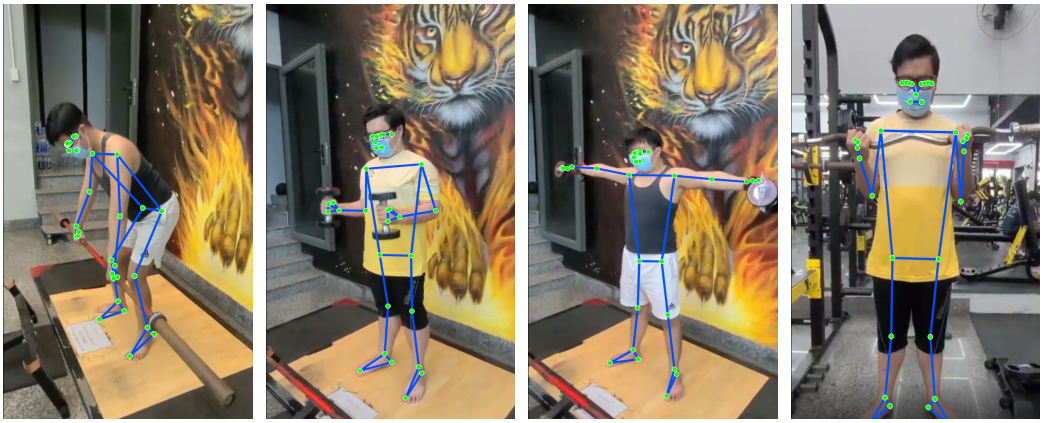
Table 1: Key specifications, provenance breakdown, and capture characteristics of the GYMED exercise dataset.

Specification / Characteristic	**Value / Distribution**
Total Videos / Total File Volume	1,024 unique video recordings / 10,225.37 MB ($\approx$ 10.2 GB)
File Formats & Video Encoding	MP4 (964 videos, 94.1%), MOV (60 videos, 5.9%) / H.264
Frame Rate & Recording Duration	30 frames per second (FPS) / 20–30 s per recording
Trimmed Action Segments	1,108 clean exercise segments (average 1.09 segments per video)
Number of Distinct Resolutions	33 resolutions (1280×720: 385; 1920×1080: 315; 720×1280: 130; others: 194)
Exercise Taxonomy Coverage	22 fine-grained resistance exercise classes
Video Partitioning Ratio (6:2:2)	Train: 580 (56.6%), Validation: 208 (20.3%), Test: 236 (23.1%)
Action Segment Partitions	Train: 639 (57.7%), Validation: 210 (19.0%), Test: 259 (23.3%)
Temporal Windows ($T = 32$ frames)	Train: 13,136 ($S = 16$), Validation: 2,075 ($S = 32$), Test: 2,743 ($S = 32$)
Multi-Source Data Provenance	Abdillah (2023) [35]: 652; YouTube: 103; Pexels: 65; Freepik: 76; Authors: 128
Subjects & Ethical Compliance	Informed consent under Declaration of Helsinki for recorded clips; diverse online athletes
Camera Dynamics & Perspectives	Static tripod and dynamic moving cameras; frontal, sagittal, oblique angles
Recording Environments	Commercial gymnasiums, domestic home workout spaces, outdoor fitness zones

graphic annotations, ensuring strictly posture-driven modeling. Furthermore, video-level partitioning guarantees that all segments and temporal windows from a given source video reside entirely within a single split, guaranteeing zero source-video overlap and eliminating temporal window leakage across training, validation, and test sets. For author-recorded videos, written informed consent was formally obtained from all participants prior to filming in strict adherence to institutional ethical standards and the Declaration of Helsinki. For web-sourced data, collection complied strictly with platform Terms of Service without harvesting private personal identifiers.

Source Statistics and Video Formats.. As summarized in Table 1, the 1,024 video corpus totals 10,225.37 MB ($\approx$ 10.2 GB) in storage, distributed between MP4 (964 videos, 94.1%) and MOV (60 videos, 5.9%) containers using H.264 compression. All videos operate at 30 frames per second. Video resolution was verified via `ffprobe`, encompassing 33 distinct resolutions dominated by standard high-definition formats: 1280×720 (385 videos, 37.6%), 1920×1080 (315 videos, 30.8%), and portrait mobile video 720×1280 (130 videos, 12.7%), with 35 videos at 854×480, 28 at 640×360, 23 at 3840×2160 (4K), and 108 videos distributed across 27 minor resolutions ($\leq$ 20 videos each).

Data Licensing and Artifact Availability.. To respect the intellectual property and terms of external video hosting platforms (YouTube, Pexels, and Freepik), original raw third-party video recordings are not redistributed. Their original source identifiers, attribution information, and manual segmentation boundaries are retained within the metadata manifests. Instead, to guarantee 100% scientific reproducibility, all derived scientific artifacts—



(a) Deadlift (b) Hammer curl (c) Lateral raise (d) Barbell biceps curl

Figure 3: Full-body 3D pose landmark detection results across representative exercise video frames using Google MediaPipe Pose: (a) Deadlift, (b) Hammer curl, (c) Lateral raise, and (d) Barbell biceps curl.

including extracted 3D skeletal landmark coordinate tensors, manual action segmentation boundaries, preprocessed kinematic features, and metadata manifests—are publicly released under the Creative Commons Attribution 4.0 International license (CC BY 4.0). All neural network architectures, training routines, and pre-trained checkpoints are openly provided under the permissive MIT License at `https://huggingface.co/Cuong2004/gym-exercise-classification`. xem lai phan nay

Video Inclusion Criteria.. To ensure high-quality skeletal extraction and biomechanically meaningful training samples, we applied the following curation protocol. A video is selected if it simultaneously satisfies four conditions: (i) it depicts a clearly identifiable single athlete performing one of the 22 target exercises; (ii) it is recorded at a frame rate of greater than or equal to 25 frames per second; (iii) it contains at least one complete repetition cycle with visible concentric and eccentric phases; and (iv) it maintains sufficient camera framing such that hip and shoulder joints remain visible for $\geq 80\%$ of the active exercise duration.

3.2. Action Trimming, Video-Level Partitioning, and Class Distribution

Raw workout recordings frequently contain non-exercise transitional intervals, such as picking up barbells, adjusting bench pins, chalking hands, and resting. To eliminate noise, all 1,024 videos were manually verified and

trimmed into action-only segments via timestamp boundary files (‘.txt‘), yielding 1,108 clean segments total (1.09 segments per video on average).

Strict Video-Level Partitioning Protocol (No Source-Video Overlap Across Splits).. To ensure rigorous evaluation without data leakage, we enforce a strict video-level partition in a 6:2:2 ratio. Our data processing follows a strictly segregated 4-step pipeline: (1) *Video-Level Isolation*: Raw video recordings are partitioned strictly by source video file into Train ($N = 580$), Validation ($N = 208$), and Test ($N = 236$) sets, ensuring no source video, derived segment, or temporal window crosses partition boundaries. (2) *Independent Landmark Extraction*: Monocular 3D skeletal landmarks are extracted independently per video file using Google MediaPipe Pose without inter-video context. (3) *Within-Video Sliding Window Segmentation*: Temporal windows ($T = 32$ frames) are extracted strictly within individual trimmed action boundaries: $S = 16$ frames (50% overlap) for Train (13,136 windows) to provide dense temporal coverage, and $S = 32$ frames (non-overlapping) for Validation (2,075 windows) and Test (2,743 windows across 233 videos with valid segments ≥ 32 frames). No sliding window is permitted to cross video or segment boundaries. (4) *Strict Training-Partition Feature Standardization*: Joint spatial coordinates are centered relative to the mid-hip origin ($p_{\text{hip_mid}} = \frac{1}{2}(p_{\text{left_hip}} + p_{\text{right_hip}})$) per frame, establishing translation invariance. To scale features uniformly without data leakage, global feature-wise z-score normalization statistics ($\mu_{\text{train}}, \sigma_{\text{train}}$) are computed strictly on the training partition and frozen, then applied to normalize training, validation, and test representations (with on-the-fly physical augmentations strictly preceding standardization). This protocol ensures zero source-video overlap, zero temporal window leakage, and zero test-information contamination in feature normalization. Because web-sourced exercise recordings lack explicit biometric subject identifiers, this strict video-level partition guarantees that all derived segments, temporal sliding windows, and repetitions from any single source video reside exclusively within a single partition (Train, Validation, or Test), eliminating temporal sequence leakage and video-level background memorization.

Class Imbalance.. Table 2 details the distribution of videos and action segments across all 22 classes. As shown, natural gym workout data exhibits significant class imbalance: highly popular compound movements such as *barbell biceps curl* ($N = 84$ videos) and *bench press* ($N = 84$ videos) pos-

Table 2: Video and action segment distributions across the 22 resistance training exercises by partition split.

Exercise Class	**Videos**				**Action Segments**			
	Train	**Val**	**Test**	**Total**	**Train**	**Val**	**Test**	**Total**
Barbell biceps curl	52	18	14	84	53	18	14	85
Bench press	51	18	15	84	51	18	15	84
Chest fly machine	28	9	8	45	29	9	8	46
Deadlift	30	10	10	50	30	10	10	50
Decline bench press	18	5	9	32	29	5	10	44
Hammer curl	24	8	14	46	28	8	14	50
Hip thrust	14	6	9	29	19	6	15	40
Incline bench press	27	10	9	46	27	10	9	46
Lat pulldown	42	15	14	71	42	15	15	72
Lateral raise	24	8	15	47	24	8	15	47
Leg extension	22	9	13	44	27	9	13	49
Leg raises	15	6	11	32	15	6	11	32
Plank	5	2	2	9	5	2	2	9
Pull-up	31	11	10	52	31	11	10	52
Push-up	40	14	12	66	43	14	14	71
Romanian deadlift	12	6	6	24	13	6	8	27
Russian twist	15	6	6	27	19	6	6	31
Shoulder press	27	10	13	50	28	10	14	52
Squat	24	10	15	49	27	11	17	55
T-bar row	15	4	10	29	17	4	10	31
Tricep pushdown	39	14	12	65	39	14	12	65
Tricep dips	25	9	9	43	43	10	25	78
Total	**580**	**208**	**236**	**1,024**	**639**	**210**	**259**	**1,108**

sess large sample representations, whereas specialized exercises such as *plank* ($N = 9$ videos total: 5 train, 2 val, 2 test) are relatively scarce.

3.3. Landmark Extraction and Anatomical Joint Selection

Every video frame is processed through Google MediaPipe Pose Heavy (complexity level 2) [8, 9], extracting 33 full-body 3D landmarks $(x_i, y_i, z_i, \text{vis}_i)$ normalized relative to bounding box and depth dimensions, as illustrated on representative workout frames in Fig. 3. Because facial landmarks and fine-grained finger points contribute negligible kinetic leverage to compound resistance exercises while introducing tracking jitter, we filter the skeleton

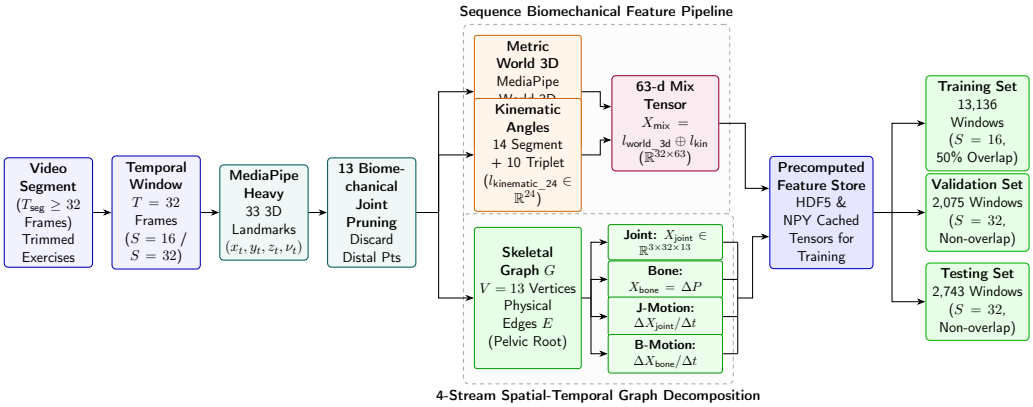


Figure 4: The proposed Biomechanical Feature Extraction and Kinematic Tensor Decomposition pipeline: illustrating the two-branch extraction of the 63-d sequence mix and the 4-stream spatial-temporal graph tensors.

to $V = 13$ major anatomical joints (detailed in Table A.17): nose (cranial orientation tracking), left/right shoulders, left/right elbows, left/right wrists, left/right hips (pelvic centroid / origin anchor), left/right knees, and left/right ankles.

4. Methodology

The overall architectural workflow of **SkelGym** is illustrated in Fig. 1, while Fig. 4 details the underlying biomechanical feature extraction and kinematic tensor decomposition. Below, we formally derive each mathematical component of our framework.

4.1. Problem Formulation and Mathematical Notation

Let $\mathcal{V} = \{I_t\}_{t=1}^{N_V}$ represent a trimmed resistance workout video of N_V consecutive RGB frames, where $I_t \in \mathbb{R}^{H\times W\times 3}$. A monocular pose estimation operator $\Phi_{\text{pose}} : I_t \mapsto \mathcal{P}_t$ extracts $V_0 = 33$ three-dimensional body landmarks $\mathcal{P}_t = \{(x_{t,v}, y_{t,v}, z_{t,v}, \nu_{t,v})\}_{v=1}^{V_0}$, where $(x, y, z) \in [-1, 1]^3$ represent normalized spatial coordinates and $\nu \in [0, 1]$ is a visibility confidence indicator.

To eliminate tracking jitter and high-frequency noise from distal extremities (such as facial landmarks and toes) that contribute negligible physical leverage to compound resistance movements, we apply an anatomical pruning operator $\Pi_{13} : \mathbb{R}^{V_0\times 4} \mapsto \mathbb{R}^{V\times 3}$ retaining $V = 13$ primary structural joints

(see Appendix Appendix A): the nose (cranial orientation tracking), bilateral shoulders, elbows, wrists, hips (pelvic centroid origin anchor), knees, and ankles.

For each action segment, temporal sampling generates fixed-length windows $w_k \in \mathbb{R}^{T \times V \times 3}$ spanning $T = 32$ frames (≈ 1.07 seconds at 30 FPS). Let $\mathcal{Y} = \{1, \ldots, C\}$ denote the label space corresponding to the $C = 22$ gym exercise categories. The primary learning objective is to parameterize a predictive distribution $P_\Theta(y = c \mid w)$ that maximizes the likelihood of the true exercise class $y \in \mathcal{Y}$, and subsequently infer a consensus video-level decision $\hat{y}_{\text{vid}}$ across all K_v constituent windows of video $\mathcal{V}$.

4.2. Biomechanical Compound Representation (63-d Kinematic Mix v2)

Directly feeding raw screen-space coordinates into neural classifiers forces models to memorize camera distance, subject translation, and 2D perspective foreshortening. Conversely, unnormalized relative coordinates discard absolute physical scale, while computing all combinatorial three-joint triplet angles ($\binom{13}{3} = 286$) induces severe multicollinearity, gradient singularities, and dimensional redundancy. To resolve this fundamental trade-off, we formulate the **Biomechanical Compound Representation (Mix v2)** ($l_{\text{mix}} \in \mathbb{R}^{63}$) by coupling pelvic-centered metric 3D World coordinates with a pruned set of 24 anatomically decisive kinematic angles:

4.2.1. Metric World 3D Coordinates ($\mathbf{1}_{world_3d}$):

We utilize MediaPipe Pose 3D World landmark trajectories $\mathcal{P}_t^{\text{world}} \in \mathbb{R}^{13 \times 3}$, which predict keypoint positions in estimated metric dimensions (meters) relative to a real-world scale origin located near the pelvic midpoint. SkelGym retains the 13 selected structural XYZ coordinates directly:

$$l_{\text{world_3d}}(t) = \left[(p_{t,1}^{\text{world}})^T, (p_{t,2}^{\text{world}})^T, \ldots, (p_{t,13}^{\text{world}})^T \right]^T \in \mathbb{R}^{39}. \tag{1}$$

Global feature-wise z-score normalization statistics ($\mu_{\text{train}}, \sigma_{\text{train}}$) are computed strictly on the training partition and applied uniformly across splits. Crucially, retaining estimated metric scale allows models to directly perceive estimated metric limb geometry and joint excursion.

4.2.2. Kinematic and Anatomical Joint/Limb Angles ($\mathbf{1}_{kinematic_24}$):

Rather than extracting combinatorial pairs ($\binom{13}{2} = 78$) or all triplets ($\binom{13}{3} = 286$) that introduce severe multicollinearity, we distill a compact set

of **24 anatomically decisive kinematic angles** specifically tailored for resistance exercise dynamics:

1. **14 Limb Segment Elevation/Orientation Angles ($\theta \in [-\pi/2, \pi/2]$):** Vertical inclinations relative to the horizontal ground plane (X–Z) for the 14 major kinetic and postural segments: bilateral upper arms (shoulder $\rightarrow$ elbow), bilateral forearms (elbow $\rightarrow$ wrist), bilateral thighs (hip $\rightarrow$ knee), bilateral shanks (knee $\rightarrow$ ankle), shoulder girdle (left $\rightarrow$ right shoulder), pelvic girdle (left $\rightarrow$ right hip), bilateral torso flanks (shoulder $\rightarrow$ hip), spine axis (mid-hip $\rightarrow$ mid-shoulder), and neck axis (mid-shoulder $\rightarrow$ nose).
2. **10 Joint Flexion/Extension Triplet Angles ($\psi \in [0, \pi]$):** 3D articulation angles at the primary load-bearing joints capturing mechanical work and posture: bilateral knees ($\angle(\text{hip}, \text{knee}, \text{ankle})$), bilateral elbows ($\angle(\text{shoulder}, \text{elbow}, \text{wrist})$), bilateral hips ($\angle(\text{shoulder}, \text{hip}, \text{knee})$), bilateral shoulders ($\angle(\text{hip}, \text{shoulder}, \text{elbow})$), torso posture ($\angle(\text{nose}, \text{mid-shoulder}, \text{mid-hip})$), and arm-splay angle ($\angle(\text{left elbow}, \text{mid-shoulder}, \text{right elbow})$).

By restricting segment orientations to signed elevations relative to gravity ($\theta = \arctan2(\Delta y, \sqrt{\Delta x^2 + \Delta z^2})$), these angles remain intrinsically invariant to horizontal camera yaw rotation while achieving a 69.2% reduction in angular dimensionality (78 $\rightarrow$ 24), completely eliminating gradient singularities at joint lockout and preventing overparameterization:

$$l_{\text{kinematic_24}}(t) = [\theta_1(t), \ldots, \theta_{14}(t),\ \psi_1(t), \ldots, \psi_{10}(t)]^T \in \mathbb{R}^{24}. \tag{2}$$

4.2.3. Compound Fusion Tensor ($\mathbf{X}_{mix}$) and Dual-Branch Projection:

The combined biomechanical descriptor at frame t concatenates the 39-d metric World 3D coordinates and 24-d kinematic angles into a 63-dimensional compound feature:

$$l_{\text{mix}}(t) = l_{\text{world_3d}}(t) \oplus l_{\text{kinematic_24}}(t) \in \mathbb{R}^{39+24} = \mathbb{R}^{63}. \tag{3}$$

Over the sliding window of $T = 32$ frames, the sequence input tensor is structured as $X_{\text{mix}} \in \mathbb{R}^{32\times 63}$. In our sequence Transformer, a **Dual-Branch Input Projector** maps the spatial coordinates (39 $\rightarrow$ 56) and kinematic angles (24 $\rightarrow$ 56) through dedicated linear transformations before concatenating them into the unified token embedding of dimension $d_{\text{model}} = 112$.

4.3. Task-Oriented Skeletal Augmentation Configuration (SkelGym-Aug)

Training deep neural networks on skeletal trajectories introduces overfitting risks due to athlete-specific anthropometry, execution styles, and camera viewing orientations. While data augmentation is standard practice in general action recognition, applying unconstrained transformations to resistance exercises risks violating anatomical bone lengths, altering gravitational loading directions, or distorting repetition tempos. To address this, we formulate a disciplined candidate pool across geometric, temporal, and noise-based domains, and determine the optimal configuration through empirical Leave-One-Out (LOO) and isolated ablations evaluated strictly on validation-window metrics.

4.3.1. Candidate Transformation Pool

The candidate augmentation pool spans five operators across three physical domains:

1. Geometric Transformations..

- **Sagittal Bilateral Reflection ($\mathbb{Z}_2$ Symmetry):** Human anatomy exhibits reflective bilateral symmetry across the sagittal plane. Let $\sigma \in \mathbb{Z}_2$ denote the sagittal reflection operator acting on spatial coordinates: $\sigma(x, y, z) = (-x, y, z)$. Bilateral anatomical joints are permuted via an exact permutation involution $\pi \in \mathfrak{S}_{13}$ that swaps contralateral counterparts while preserving axial medial keypoints:

$$\pi = (0)(1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)(11\ 12), \tag{4}$$

  corresponding to nose ($0 \leftrightarrow 0$), shoulders ($1 \leftrightarrow 2$), elbows ($3 \leftrightarrow 4$), wrists ($5 \leftrightarrow 6$), hips ($7 \leftrightarrow 8$), knees ($9 \leftrightarrow 10$), and ankles ($11 \leftrightarrow 12$). Sagittal reflection acts as an improper Euclidean isometry ($\det = -1$) in $E(3)$ that doubles effective training diversity while strictly preserving metric bone lengths and joint angles.

- **Gravitational Yaw Rotation ($\mathrm{SO}(2)$ Invariance):** To simulate horizontal camera viewpoints around the athlete without altering the gravitational vertical axis y, coordinates are rotated about the vertical axis by continuous random angle $\alpha \sim \mathcal{U}(-15°, +15°)$ using the planar rotation matrix $R_y(\alpha) \in \mathrm{SO}(2) \subset \mathrm{SO}(3)$:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \tag{5}$$

Because $R_y(\alpha)$ acts strictly within the transverse plane, vertical displacement remains invariant ($\Delta y' = \Delta y$) and horizontal Euclidean radius is identically preserved:

$$\Delta x'^2 + \Delta z'^2 = (\Delta x \cos\alpha + \Delta z \sin\alpha)^2 + (-\Delta x \sin\alpha + \Delta z \cos\alpha)^2 = \Delta x^2 + \Delta z^2. \tag{6}$$

Consequently, elevation angles satisfy $\theta' = \theta$, preserving biological limb inclination relative to gravity while teaching the network horizontal viewpoint invariance.

- **Spatial Proportional Scaling:** Isotropic scaling $s \sim \mathcal{U}(0.90, 1.10)$ applied to $\tilde{p}$ to evaluate synthetic stature variation.

2. Temporal and Noise-Based Perturbations..

- **Temporal Time Warping:** Execution speed perturbation via linear temporal resampling ($s_t \sim \mathcal{U}(0.85, 1.15)$) followed by uniform interpolation back to $T = 32$ frames.
- **Gaussian Sensor Jitter:** Coordinate jitter $\epsilon \sim \mathcal{N}(0, \sigma^2 I_3)$ with $\sigma = 0.008$ injected into physical coordinates to simulate high-frequency tracking noise.

4.3.2. Empirical Selection and Final SkelGym-Aug Configuration

Evaluating candidate transformations on validation-window metrics (detailed in Section 5.4.2) reveals that combining all five operators degrades representation learning (Validation Window Macro F1 drops to 0.7882). Synthetic scaling perturbs absolute metric bone dimensions, sensor jitter corrupts velocity derivatives, and temporal warping disrupts phase duration ratios (concentric/eccentric cadence). Conversely, the paired configuration of **Sagittal Bilateral Reflection + Gravitational Yaw Rotation** strictly preserves pairwise Euclidean distances and gravitational verticality, achieving peak representation learning across all 13 experimental configurations tested (Validation Window Accuracy: **81.20% ± 0.72%**, Macro F1: **0.8119 ± 0.0072**, a decisive gain of +2.34% F1 over unaugmented clean baseline). Therefore, the **final SkelGym-Aug configuration** is established as the 2-operator suite: *Sagittal Bilateral Reflection + Gravitational Yaw Rotation (±15°)*.

Zero Test-Time Data Corruption Protocol.. All transformations within the SkelGym-Aug pipeline are executed *strictly on-the-fly during training forward passes* as stochastic data regularization. Validation and held-out test splits remain completely pristine and unaugmented throughout model selection, hyperparameter tuning, and final evaluation, ensuring zero test-time corruption and guaranteeing that reported metrics reflect authentic generalization to unmanipulated video recordings.

4.4. Self-Attention Sequence Modeling (Transformer Mix v2)

The augmented sequence tensor $X_{\text{mix}} \in \mathbb{R}^{32\times 63}$ is processed by a temporal Transformer encoder [14] calibrated to an efficient parameter budget of $\approx$ 301K parameters to ensure strictly fair capacity parity with recurrent backbones. The architecture consists of four sequential processing stages: dual-branch projection, temporal positional encoding, stacked multi-head self-attention with feed-forward sub-layers, and global temporal pooling.

Dual-Branch Embedding and Positional Encoding.. At each discrete time step $t \in \{1, \ldots, 32\}$, the 63-dimensional compound feature vector is routed through parallel coordinate and angular linear projection branches:

$$e_{\text{coord}}(t) = l_{\text{world_3d}}(t)W_{\text{coord}} + b_{\text{coord}} \in \mathbb{R}^{56}, \quad e_{\text{angle}}(t) = l_{\text{kinematic_24}}(t)W_{\text{angle}} + b_{\text{angle}} \in \mathbb{R}^{56}, \tag{7}$$

where $W_{\text{coord}} \in \mathbb{R}^{39\times 56}$ and $W_{\text{angle}} \in \mathbb{R}^{24\times 56}$. Concatenating these branch outputs forms a balanced latent embedding token of dimension $d_{\text{model}} = 112$:

$$E_0(t) = [e_{\text{coord}}(t) \oplus e_{\text{angle}}(t)] \in \mathbb{R}^{112}. \tag{8}$$

Because the core self-attention operator is permutation-invariant, sequence chronology is explicitly preserved by superimposing learnable temporal positional embeddings $E_{\text{pos}} \in \mathbb{R}^{32\times 112}$ initialized via a normal distribution ($\sigma = 0.02$):

$$Z^{(0)} = E_0 + E_{\text{pos}} \in \mathbb{R}^{32\times 112}. \tag{9}$$

This formulation enables the model to adaptively capture phase-dependent kinematic progressions throughout repetition cycles.

Multi-Head Self-Attention.. The embedded sequence is propagated through $L = 3$ identical Transformer encoder layers. Each layer incorporates a multi-head self-attention (MHSA) module configured with $h = 4$ parallel attention heads ($d_k = d_v = 28$, $d_{\text{ff}} = 168$). For head $j \in \{1, \ldots, h\}$, queries, keys, and

values are generated via linear projections $Q_j = ZW_{Q,j}$, $K_j = ZW_{K,j}$, and $V_j = ZW_{V,j}$:

$$\text{head}_j = \text{Softmax}\left(\frac{Q_j K_j^T}{\sqrt{d_k}}\right) V_j, \tag{10}$$

$$\text{MHSA}(Z) = [\text{head}_1 \oplus \cdots \oplus \text{head}_h]\, W_O, \tag{11}$$

where $W_O \in \mathbb{R}^{112\times 112}$ aggregates the concatenated multi-head outputs into the model embedding dimension.

Residual Normalization and Feed-Forward Blocks.. To ensure stable gradient propagation and prevent representation collapse, each sub-layer employs residual skip connections coupled with pre-layer normalization:

$$Z'^{(l)} = Z^{(l-1)} + \text{MHSA}\left(\text{LayerNorm}(Z^{(l-1)})\right), \tag{12}$$

$$Z^{(l)} = Z'^{(l)} + \text{FFN}\left(\text{LayerNorm}(Z'^{(l)})\right). \tag{13}$$

Here, the position-wise feed-forward network is parameterized as $\text{FFN}(u) = \text{GELU}(uW_1 + b_1)W_2 + b_2$ using Gaussian Error Linear Units [25], with an expansion hidden dimension of $d_{\text{ff}} = 168$ and dropout rate of 0.2.

Global Pooling and Classification Head.. Following the final layer L, the frame representations are temporally summarized via uniform average pooling, $\hat{z} = \frac{1}{T}\sum_{t=1}^{T} Z_t^{(L)} \in \mathbb{R}^{112}$, and projected to normalized class posterior probabilities:

$$\hat{P}_{\text{trans}} = \text{Softmax}\left(\hat{z}W_{\text{cls}} + b_{\text{cls}}\right) \in \Delta^{21}. \tag{14}$$

4.5. Multi-Stream Spatial-Temporal Adaptive Graph Convolution (4-Stream AAGCN)

While Transformers model global sequence dynamics, physical musculoskeletal connectivity is naturally represented as a spatio-temporal graph $G = (V, E)$ [16], where vertices $V = \{1, \ldots, 13\}$ denote joints and edges E denote physical bones. To capture rich kinematic perspectives, we formulate four parallel input streams following multi-stream principles [17, 18]:

$$X_{\text{joint}}(c,t,v) = \tilde{p}_{t,v}^{(c)} \in \mathbb{R}^{3\times T\times V}, \tag{15}$$

$$X_{\text{bone}}(c,t,v) = \tilde{p}_{t,v_{\text{target}}}^{(c)} - \tilde{p}_{t,v_{\text{source}}}^{(c)} \in \mathbb{R}^{3\times T\times V}, \tag{16}$$

$$X_{\text{j-motion}}(c,t,v) = X_{\text{joint}}(c,t+1,v) - X_{\text{joint}}(c,t,v) \in \mathbb{R}^{3\times (T-1)\times V}, \tag{17}$$

$$X_{\text{b-motion}}(c,t,v) = X_{\text{bone}}(c,t+1,v) - X_{\text{bone}}(c,t,v) \in \mathbb{R}^{3\times (T-1)\times V}, \tag{18}$$

where $(v_{\text{source}}, v_{\text{target}})$ represents directed bone edges originating from the pelvic center.

Each stream is independently processed by a lightweight calibrated Adaptive Graph Convolutional Network (AAGCN) [18]. For a given skeletal representation $X \in \mathbb{R}^{B\times C\times T\times V}$, the spatial graph convolution is formulated as:

$$H_{\text{spatial}} = \sigma\left(\text{BN}\left(\left(\left(A_{\text{topo}} \odot M\right) + B + C(X)\right) XW\right)\right), \tag{19}$$

where spatial connectivity is modulated by three complementary components:

1. $A_{\text{topo}} \in \mathbb{R}^{V\times V}$ is the normalized static anatomical adjacency matrix with self-loops ($\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$) representing physical musculoskeletal bone connections, modulated element-wise by a learnable edge importance mask $M \in \mathbb{R}^{V\times V}$.
2. $B \in \mathbb{R}^{V\times V}$ is a global learnable continuous dependency matrix initialized with small Gaussian weights ($\sigma = 10^{-4}$), enabling the network to discover non-local joint correlations (e.g., wrist-ankle synchronization during deadlifts).
3. $C(X) \in \mathbb{R}^{V\times V}$ is an instance-specific, dynamic graph attention matrix computed via normalized dot-product attention over temporally pooled joint representations:

$$C(X) = \text{Softmax}\left(\frac{\theta(X)^T \phi(X)}{\sqrt{d_e}}\right), \tag{20}$$

   where $\theta(X), \phi(X) \in \mathbb{R}^{V\times d_e}$ project pooled joint features into an embedding space of dimension $d_e = 16$.

Spatial graph convolution is paired with a temporal convolution (9×1 temporal kernel, padding 4), batch normalization, dropout (0.2), and residual shortcut connections with GELU activation. The backbone is structured into a calibrated 3-stage hierarchy with channel capacities $[48, 96, 150]$ and temporal strides $[1, 1, 2]$, followed by global average pooling and a two-stage classification head, constraining the parameter footprint to $\approx$ 378K parameters per stream.

4.6. Cross-Paradigm Ensemble Fusion

The SkelGym framework unifies two fundamentally distinct, complementary learning paradigms: the Sequence Self-Attention Model M_{seq} (Transformer Mix v2) and the Spatial-Temporal Graph Kinematics Model M_{graph}

(AAGCN). Within the graph paradigm, the four kinematic streams (Joint, Bone, Joint-Motion, Bone-Motion) capture orthogonal spatial and velocity characteristics. They are unified via late probability averaging:

$$P_{\text{graph}}(w) = \frac{1}{4} \sum_{k \in \{\text{joint,bone,j-mot,b-mot}\}} P_k(w). \tag{21}$$

To address diverse edge computing constraints, we establish two canonical deployment tiers:

1. **SkelGym-Lite (Edge-Efficient Two-Model Ensemble):** Pairs the sequence Transformer (M_{seq}) with the single strongest individual graph stream (AAGCN Bone, $M_{\text{graph}}^{\text{bone}}$). Voting probabilities are aggregated via *Accuracy-Weighted Soft Voting*, where stream weights are proportional to validation window accuracy:

$$P_{\text{lite}}(w) = w_{\text{trans}} P_{\text{trans}}(w) + w_{\text{bone}} P_{\text{bone}}(w), \quad w_m = \frac{\text{Acc}_m^{\text{val}}}{\sum_j \text{Acc}_j^{\text{val}}}. \tag{22}$$

   This edge tier provides a lightweight footprint of only 679K parameters and 221.72 MFLOPs.
2. **SkelGym-Full (Canonical Cross-Paradigm Ensemble):** Integrates all five complementary backbones (Transformer Mix + Joint + Bone + Joint-Motion + Bone-Motion). Rather than tuning complex parameterized meta-classifiers or constrained non-linear optimization weights that risk overfitting validation partitions, SkelGym-Full employs **Zero-Parameter Uniform Average Soft Voting**:

$$P_{\text{full}}(w) = \frac{1}{5} \left(P_{\text{trans}}(w) + P_{\text{joint}}(w) + P_{\text{bone}}(w) + P_{\text{j-mot}}(w) + P_{\text{b-mot}}(w) \right). \tag{23}$$

Because sequence self-attention models temporal phase transitions across combined spatial-angular coordinates while graph convolutions model topological limb kinetics, their posterior errors are largely decorrelated. Uniform soft voting provides a robust, zero-parameter integration that completely avoids optimization bias on validation splits and enforces strict post-validation test separation.

4.7. Temporal Video Consensus Aggregation

For real-world continuous workout monitoring, individual 1-second window predictions ($w_{v,k}$) are aggregated across all K_v constituent temporal windows of video v via expectation-based temporal consensus pooling:

$$\hat{P}_{\text{vid}}(c) = \frac{1}{K_v} \sum_{k=1}^{K_v} P_c(w_{v,k}), \quad \hat{y}_{\text{vid}} = \arg \max_{c \in \{1,\dots,C\}} \hat{P}_{\text{vid}}(c). \tag{24}$$

Temporal consensus acts as an effective low-pass posterior probability filter, cancelling out isolated frame tracking dropouts, boundary transition noise (such as un-racking weights or pausing between repetitions), and momentary camera occlusions.

5. Experimental Results

5.1. Implementation and Reproducibility Configuration

All models were implemented in PyTorch 2.x [30] and trained across three independent random seeds (seed $\in \{42, 123, 3407\}$); within each trial, the assigned seed was fixed across data loaders, weight initialization, and GPU operations to ensure deterministic experimental repeatability. Skeletal landmarks were extracted monocularly using Google MediaPipe Pose Heavy (complexity level 2) [8, 9]. Sequences were standardized to $T = 32$ frames ($\approx$ 1.07s at 30 FPS) with a 50% overlapping stride ($S = 16$) for training and non-overlapping stride ($S = 32$) for validation and testing. Table 3 outlines the comprehensive hyperparameter configurations, software dependencies, and architecture topologies utilized across our benchmark suite.

To ensure fair scientific comparison, all individual model backbones were constrained to an identical parameter footprint of $\approx 350\text{K} \pm 15\%$ parameters: Transformer Mix v2 (301K), AAGCN (378K per stream), BiLSTM (360K), LSTM (362K), and ST-GCN (350K). All models were trained for up to 100 epochs using AdamW [24] with weight decay 10^{-4}, Cosine Annealing learning rate scheduling (with 5 warmup epochs down to a minimum learning rate of 10^{-6}), batch size of 16 (for sequence models) or 32 (for graph models), an early stopping patience of 10 epochs monitoring validation Macro F1, label smoothing regularization [26] ($\alpha = 0.05$), and Automatic Mixed Precision (AMP). Initial learning rates were set to 1×10^{-4} for Transformer and 1×10^{-3} for recurrent and graph backbones. Evaluation metrics were computed using Scikit-learn [33].

Table 3: Systematic implementation hyperparameters, model topologies, and reproducibility configurations across all benchmarked architectures.

Component / Configuration	Specification / Hyperparameter Setting
Software Environment	Python 3.10+, PyTorch 2.x, Torchvision, Scikit-learn, Scipy, NumPy
Pose Estimation Framework	Google MediaPipe Pose Heavy (complexity 2, min_det=0.5, min_track=0.5)
Hardware Infrastructure	Apple Silicon M4 (MPS & CPU), NVIDIA RTX PRO 6000 Blackwell (CUDA)
Random Seeds	3 independent runs (seed $\in \{42, 123, 3407\}$) fixed across data loaders, initializers, and ops (canonical single-run seed 42)
Temporal Window Length	$T = 32$ frames (≈ 1.07 s at 30 FPS)
Sliding Window Strides	Train: $S = 16$ frames (50% overlap); Validation and Test: $S = 32$ frames
Missing Frame Imputation	Linear coordinate interpolation across contiguous valid joint detections
Feature Normalization	Hip midpoint centering ($p_i - p_{\text{hip_mid}}$) + Train-set z-score standardization
Optimizer & Weight Decay	AdamW [24] ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$)
Learning Rate Schedule	Cosine Annealing with 5 warmup epochs down to $\eta_{\min} = 10^{-6}$
Initial Learning Rates	Transformer: 1×10^{-4}; AAGCN / ST-GCN / LSTM / BiLSTM: 1×10^{-3}
Batch Size & Max Epochs	Batch size: 16 (Sequence models), 32 (Graph models); Maximum epochs: 100
Early Stopping Criterion	Patience = 10 epochs monitoring validation Macro F1 score
Loss Function & Smoothing	Cross-Entropy Loss with label smoothing $\alpha = 0.05$ (optional Focal Loss $\gamma = 2.0$)
Precision Acceleration	PyTorch Automatic Mixed Precision (AMP) with `bfloat16` / `float16`
Transformer Architecture	3 layers, 4 attention heads, $d_{\text{model}} = 112$, $d_{\text{ff}} = 168$, learnable PE, dropout 0.2 (301K params)
AAGCN Graph Architecture	3 stages [48, 96, 150], kernel 9, adaptive residual topology (378K params/stream)
SkelGym-Aug Parameters	Bilateral body reflection ($p = 0.5$); Gravitational 3D yaw $\alpha \in [-15°, +15°]$

Custom Lightweight Architecture Design and Training from Scratch.. Crucially, none of our models rely on massive pre-trained vision-language backbones, heavy transfer-learned weights (e.g., from NTU RGB+D or ImageNet), or external video foundation models. All architectures were custom-engineered from the ground up to operate within an ultra-compact parameter footprint ($\approx 350\text{K} \pm 15\%$ parameters). Every model was trained strictly from scratch directly on the SkelGym training split using random weight initialization: He/Kaiming normal initialization [31] for convolutional and linear projection layers, and orthogonal initialization for recurrent state matrices. This deliberate architectural design constraint eliminates external dataset dependencies, guarantees fully transparent and reproducible convergence dynamics, and maintains ultra-low computational latency (0.49–3.96 ms on host CPU, 0.59–4.65 ms on CUDA, 1.36–5.86 ms on MPS, where individual backbones require 0.49–1.03 ms) essential for real-time edge processing on consumer smartphones and wearable micro-controllers.

5.2. Master Benchmark Evaluation across Baselines and Proposed Methods

Table 4 presents the comprehensive master benchmark comparing single-window inference ($T = 32$ frames, ≈ 1.07s) against temporal video consensus across all baseline models, external architectures, and our proposed SkelGym frameworks. To ensure fair and rigorous scientific evaluation, all results in Table 4 are reported strictly on the held-out test partition ($N = 2,743$ test windows, $N = 233$ test videos) across three independent random seeds (seed $\in \{42, 123, 3407\}$). We structure the benchmarked models into three

Table 4: Comprehensive Master Benchmark comparison across baseline architectures and proposed SkelGym frameworks evaluated strictly on the held-out test partition ($N = 2,743$ windows, $N = 233$ videos). Results are reported as Mean ± SD across 3 independent random seeds $(42, 123, 3407)$.

Model Architecture	Feature Representation	Paradigm / Modality	Params	FLOPs	Test Win Acc (%)	Test Win F1	Test Vid Acc (%)	Test Vid F1
Category 1: Baseline Models on Default / Raw 3D Coordinates								
ST-GCN (Raw 3D Clean) [16]	Raw 3D Landmarks	Fixed Graph Convolutional	350K	184.20 M	48.35% ± 0.94%	0.4701 ± 0.0141	58.65% ± 1.76%	0.5432 ± 0.0208
BlockGCN (CVPR 2024 Adapted) [57]	33 Raw 3D Joints	Multi-Order Spatial-Temporal Graph	1.35M	212.40 M	54.24% ± 1.04%	0.5576 ± 0.0047	70.48% ± 0.40%	0.6948 ± 0.0024
LSTM (Raw 3D Coordinates)	Raw 3D Coordinates (39-d)	Recurrent Sequential	362K	11.58 M	52.77% ± 1.77%	0.5041 ± 0.0250	—	—
BiLSTM (Raw 3D Coordinates)	Raw 3D Coordinates (39-d)	Bidirectional Recurrent	360K	13.90 M	53.25% ± 0.88%	0.5120 ± 0.0087	—	—
Transformer (Raw 3D Coordinates)	Raw 3D Coordinates (39-d)	Self-Attention Sequential	301K	18.87 M	62.87% ± 0.69%	0.6202 ± 0.0097	—	—
Category 2: Sequential and Graph Backbones on Biomechanical Representations								
LSTM (Biomechanical Mix v2)	Biomechanical Mix v2 (63-d)	Recurrent Sequential	362K	11.58 M	62.80% ± 0.83%	0.6171 ± 0.0089	72.39% ± 1.76%	0.7077 ± 0.0172
BiLSTM (Biomechanical Mix v2)	Biomechanical Mix v2 (63-d)	Bidirectional Recurrent	360K	13.90 M	64.82% ± 0.37%	0.6316 ± 0.0027	75.39% ± 1.46%	0.7342 ± 0.0144
Transformer (Mix v2 Clean)	Biomechanical Mix v2 (63-d)	Self-Attention Sequential	301K	18.87 M	69.24% ± 0.20%	0.6815 ± 0.0017	79.83% ± 1.27%	0.7806 ± 0.0214
Transformer (Mix v2 + SkelGym-Aug)	Biomechanical Mix v2 (63-d)	Self-Attention Sequential	301K	18.87 M	73.14% ± 0.90%	0.7253 ± 0.0106	82.69% ± 1.66%	0.8187 ± 0.0196
AAGCN (Bone 3D + SkelGym-Aug)	Bone 3D Vector ($V = 13$)	Adaptive Graph Convolutional	378K	202.86 M	66.36% ± 1.41%	0.6548 ± 0.0173	75.39% ± 2.33%	0.7440 ± 0.0335
Four-Stream AAGCN (Uniform Soft)	4 Kinematic Streams ($V = 13$)	Unified Graph Ensemble (4 Streams)	1.51M	811.44 M	74.50% ± 0.65%	0.7341 ± 0.0048	84.41% ± 1.01%	0.8308 ± 0.0179
Category 3: Proposed Cross-Paradigm Framework (SkelGym)								
SkelGym-Lite (2 Streams)	**Transformer Mix + Bone AAGCN**	**Compact Hybrid (2 Streams)**	**679K**	**221.72 M**	**73.74% ± 0.63%**	**0.7310 ± 0.0058**	**82.12% ± 0.40%**	**0.8198 ± 0.0030**
SkelGym-Full (5 Streams)	**Transformer Mix + 4-Stream AAGCN**	**Full Cross-Paradigm (5 Streams)**	**1.81M**	**830.29 M**	**76.19% ± 0.16%**	**0.7537 ± 0.0014**	**85.27% ± 0.41%**	**0.8386 ± 0.0075**

distinct methodological categories:

1. **Category 1: Baseline Architectures on Default / Raw 3D Coordinates:** Evaluates standard sequence and graph architectures trained on raw Euclidean coordinates without coordinate recentering or engineered biomechanical encodings. This category includes baseline ST-GCN [16], the strong external baseline BlockGCN (CVPR 2024 Adapted) [57], and sequential backbones (LSTM, BiLSTM, Transformer) operating on 39-d raw Cartesian landmark coordinates.
2. **Category 2: Sequential and Graph Backbones on Biomechanical Representations:** Evaluates individual sequential and multi-stream graph backbones trained on metric World 3D and our proposed 63-d Biomechanical Mix v2 representations, both with and without data augmentation.
3. **Category 3: Proposed Cross-Paradigm Frameworks (SkelGym-Lite and SkelGym-Full):** Evaluates our multi-stream cross-paradigm ensembles that synergistically unite self-attention temporal modeling with adaptive spatial-temporal graph convolutions.

Analysis of Empirical Baseline Disparities.. As evidenced in Category 1 of Table 4, standard vision baselines trained directly on raw coordinates exhibit substantial performance degradation. Static ST-GCN [16] achieves merely 48.35% window accuracy and 58.65% video consensus accuracy (Video Macro F1: 0.5432). Because resistance training involves non-local physical interactions (e.g., wrist-ankle stabilization during deadlifts), a static anatomical adjacency graph restricts 1-hop message passing, preventing the network from

modeling compound limb coordination. Similarly, the strong external baseline BlockGCN [57] (adapted faithfully to 33 raw MediaPipe landmarks from CVPR 2024) achieves 54.24% window accuracy and 70.48% video consensus accuracy (Video Macro F1: 0.6948). While BlockGCN's multi-order graph convolutions outperform ST-GCN by +11.83% on video accuracy, its reliance on raw Cartesian coordinates leaves it vulnerable to camera distance and perspective distortion.

Efficacy of Proposed Biomechanical Representations.. Transitioning from raw coordinates to our proposed 63-dimensional Biomechanical Mix v2 representation (Category 2) yields dramatic performance gains across all model families. For sequential backbones, replacing raw 3D coordinates with Biomechanical Mix v2 elevates LSTM window accuracy from 52.77% to **62.80% ± 0.83%** (+10.03% absolute surge) and video accuracy to 72.39%, BiLSTM from 53.25% to **64.82% ± 0.37%** (+11.57% surge) and video accuracy to 75.39%, and Transformer from **62.87% ± 0.69%** to **69.24% ± 0.20%** under clean training (+6.37% absolute surge), reaching **73.14% ± 0.90%** window accuracy and **82.69% ± 1.66%** video accuracy when augmented with SkelGym-Aug.

Temporal Video Consensus Gains.. Across every evaluated architecture in Table 4, transitioning from isolated single-window predictions ($T = 32$ frames) to temporal video consensus produces a substantial performance surge, ranging from +8.59% to +10.77% in absolute test accuracy. For SkelGym-Full, temporal consensus elevates test accuracy from **76.19% ± 0.16%** up to **85.27% ± 0.41%** (+9.08% surge) and Macro F1 from 0.7537 to 0.8386, correctly classifying 200 out of 233 held-out test videos (85.84% accuracy on the canonical seed 42 run). Temporal consensus acts as an effective posterior probability filter that cancels out transient pose estimation tracking jitter, boundary transition noise (e.g., racking and un-racking weights), and brief limb occlusions.

Superiority of Proposed SkelGym Ensembles.. The proposed SkelGym-Full ensemble achieves the highest performance across all evaluated metrics: **76.19% ± 0.16%** test window accuracy and **85.27% ± 0.41%** test video consensus accuracy (Video Macro F1: **0.8386 ± 0.0075**). Compared to baseline ST-GCN, SkelGym-Full achieves an absolute advantage of **+26.62%** in video consensus accuracy; compared to Strong Baseline BlockGCN, SkelGym-Full delivers an advantage of **+14.79%** in video consensus accuracy (+0.1438

Video Macro F1), while maintaining an ultra-compact footprint of 1.81M parameters. For resource-constrained edge devices, SkelGym-Lite requires merely 679K parameters and 221.72 MFLOPs while delivering **82.12%** ± **0.40%** test video consensus accuracy, surpassing all standalone sequential backbones and rivaling complex 4-stream graph models.

5.3. Per-Class Performance Breakdown — SkelGym-Full

Table 5 presents a fine-grained per-class breakdown of the proposed SkelGym-Full ensemble (Uniform Soft Voting), reporting Precision, Recall, F1-Score, and Support independently across both the 2,743 held-out test windows and the 233 test videos (200 of 233 videos correctly classified, achieving 85.84% canonical accuracy). Fig. 5 illustrates the corresponding normalized confusion matrix. Analyzing the empirical metrics across all 22 resistance exercises reveals three distinct performance tiers:

High-Precision Distinctive Movements (F1 ≥ 0.90*)..* Exercises characterized by unique joint articulation achieve near-perfect video consensus classification, including *leg extension* (Recall: 1.0000, F1: 1.0000 across 13 videos), *lateral raise* (Recall: 1.0000, F1: 1.0000 across 15 videos), *leg raises* (Recall: 1.0000, F1: 1.0000 across 11 videos), *russian twist* (Recall: 1.0000, F1: 1.0000 across 6 videos), *squat* (Recall: 1.0000, F1: 1.0000 across 15 videos), *push-up* (Recall: 1.0000, F1: 0.9600 across 12 videos), *chest fly machine* (Recall: 1.0000, F1: 0.9412 across 8 videos), and *hip thrust* (Recall: 0.8889, F1: 0.9412 across 9 videos). These movements operate in isolated topological subspaces—such as seated tibial rotation against a static femur in leg extensions, or bilateral transverse torso rotation in Russian twists—which prevents kinematic confusion.

Robust Compound Resistance Exercises (F1 $\in [0.70, 0.90)$*)..* Movements involving standard planar flexion-extension trajectories exhibit strong, reliable recognition: *lat pulldown* (Recall: 1.0000, F1: 0.8966 across 13 videos), *tricep dips* (Recall: 0.8889, F1: 0.8889 across 9 videos), *tricep pushdown* (Recall: 0.9167, F1: 0.8462 across 12 videos), *pull-up* (Recall: 0.7000, F1: 0.8235 across 10 videos), *t-bar row* (Recall: 0.7000, F1: 0.8235 across 10 videos), *barbell biceps curl* (Recall: 1.0000, F1: 0.8000 across 14 videos), *plank* (Recall: 1.0000, F1: 0.8000 across 2 videos), *deadlift* (Recall: 0.9000, F1: 0.7826 across 10 videos), *incline bench press* (Recall: 0.7778, F1: 0.7778 across 9 videos), *decline bench press* (Recall: 0.7500, F1: 0.7500 across 8 videos), and

shoulder press (Recall: 0.6923, F1: 0.7200 across 13 videos). For these compound exercises, temporal consensus effectively resolves transitional pauses, elevating recall by 15% to 30% over isolated windows.

Kinematic Ambiguity and Fine-Grained Error Clusters (F1 $<$ 0.70).. The remaining error mass is concentrated within three tightly coupled biomechanical exercise clusters. Specifically, *hammer curl* (Video Recall: 0.5000, F1: 0.6364 across 14 videos) experiences confusion with *barbell biceps curl* (Video Recall: 1.0000, F1: 0.8000), where hammer curl test windows are misclassified as biceps curls due to identical sagittal elbow flexion profiles. Similarly, *Romanian deadlift* (Video Recall: 0.5000, F1: 0.6000 across 6 videos) exhibits mutual confusion with conventional *deadlift* (Video Recall: 0.9000, F1: 0.7826) whenever athletes perform sub-optimal knee travel. Finally, pressing angle variations within the bench press family produce directional ambiguity across flat *bench press* (Video Recall: 0.6429, F1: 0.6923 across 14 videos), *decline bench press* (Video Recall: 0.7500, F1: 0.7500), and *incline bench press* (Video Recall: 0.7778, F1: 0.7778). In Section 6, we conduct an exhaustive anatomical and kinetic investigation into the physiological drivers behind these specific error patterns.

5.4. Ablation Studies and Kinematic Modeling Analysis

To adhere strictly to zero-leakage experimental protocols, all model selection decisions, feature encodings, augmentation protocols, and fusion strategies were audited strictly on the **Validation Set** ($N = 2,075$ windows, $N = 208$ videos). In this section, we systematically analyze the empirical findings across our ablation suites.

5.4.1. Ablation of Landmark Feature Representations

Table 6 presents the complete 30 systematic feature screening runs across three sequential architectures (LSTM, BiLSTM, Dual-Branch Transformer) and ten distinct kinematic encodings, evaluated strictly on validation window metrics across 3 random seeds $(42, 123, 3407)$. Table 7 further presents the temporal architecture comparison between metric World 3D (39-d) and our proposed Biomechanical Mix v2 (63-d).

2D versus 3D Coordinates and World Metric Grounding.. Across all baseline architectures, 3D representations consistently outperform 2D counterparts. In resistance training, compound movements—such as squats, dead-

Table 5: Per-class performance breakdown of the proposed SkelGym-Full ensemble (Uniform Soft Voting) evaluated on held-out test partitions at temporal window level ($N = 2,743$) and video consensus level ($N = 233$, 200 correct videos = 85.84%).

Exercise Class	**Window Level** ($N = 2,743$)				**Video Consensus Level** ($N = 233$)			
	Precision	**Recall**	**F1-Score**	**Support**	**Precision**	**Recall**	**F1-Score**	**Support**
barbell biceps curl	0.4069	0.8551	0.5514	69	0.6667	1.0000	0.8000	14
bench press	0.4091	0.6562	0.5040	96	0.7500	0.6429	0.6923	14
chest fly machine	0.9213	1.0000	0.9591	82	0.8889	1.0000	0.9412	8
deadlift	0.4046	0.7910	0.5354	67	0.6923	0.9000	0.7826	10
decline bench press	0.6270	0.5896	0.6077	134	0.7500	0.7500	0.7500	8
hammer curl	0.5932	0.4348	0.5018	161	0.8750	0.5000	0.6364	14
hip thrust	0.9245	0.6229	0.7443	236	1.0000	0.8889	0.9412	9
incline bench press	0.7397	0.7105	0.7248	76	0.7778	0.7778	0.7778	9
lat pulldown	0.6691	0.9300	0.7782	100	0.8125	1.0000	0.8966	13
lateral raise	1.0000	0.9484	0.9735	155	1.0000	1.0000	1.0000	15
leg extension	0.9542	1.0000	0.9766	125	1.0000	1.0000	1.0000	13
leg raises	0.9898	0.8509	0.9151	114	1.0000	1.0000	1.0000	11
plank	0.7925	0.7500	0.7706	56	0.6667	1.0000	0.8000	2
pull Up	0.7971	0.6707	0.7285	82	1.0000	0.7000	0.8235	10
push-up	0.9062	1.0000	0.9508	87	0.9231	1.0000	0.9600	12
romanian deadlift	0.6792	0.4932	0.5714	146	0.7500	0.5000	0.6000	6
russian twist	0.9844	0.9197	0.9509	137	1.0000	1.0000	1.0000	6
shoulder press	0.7101	0.6447	0.6759	152	0.7500	0.6923	0.7200	13
squat	0.9384	0.8319	0.8820	238	1.0000	1.0000	1.0000	15
t bar row	0.8400	0.5385	0.6562	117	1.0000	0.7000	0.8235	10
tricep Pushdown	0.6083	0.7849	0.6854	93	0.7857	0.9167	0.8462	12
tricep dips	0.8987	0.9682	0.9322	220	0.8889	0.8889	0.8889	9
Macro Average	**0.7634**	**0.7723**	**0.7534**	**2,743**	**0.8626**	**0.8572**	**0.8491**	**233**
Overall Accuracy		**76.41%**		**2,743**		**85.84% (200/233)**		**233**

lifts, and bench presses—operate predominantly within the sagittal or transverse anatomical planes. When captured from oblique angles, monocular 2D projections collapse depth variations along the optical axis (z), inducing visual ambiguity. Utilizing MediaPipe World coordinates measured in physical meters anchors limb trajectories to estimated metric scale, providing consistent metric distances regardless of camera zoom.

Pelvic Coordinate Centering.. Subtracting the pelvic hip midpoint centroid ($p_i - p_{\text{hip_mid}}$) yields substantial performance gains across all sequence backbones. Raw coordinates tie features directly to the camera frame, forcing models to expend parameter capacity memorizing subject location and bounding-box translations. Centering all joints relative to the pelvic midpoint eliminates global translation bias and anchors the skeleton to the bodily center of mass, allowing the network to focus purely on relative joint dynamics.

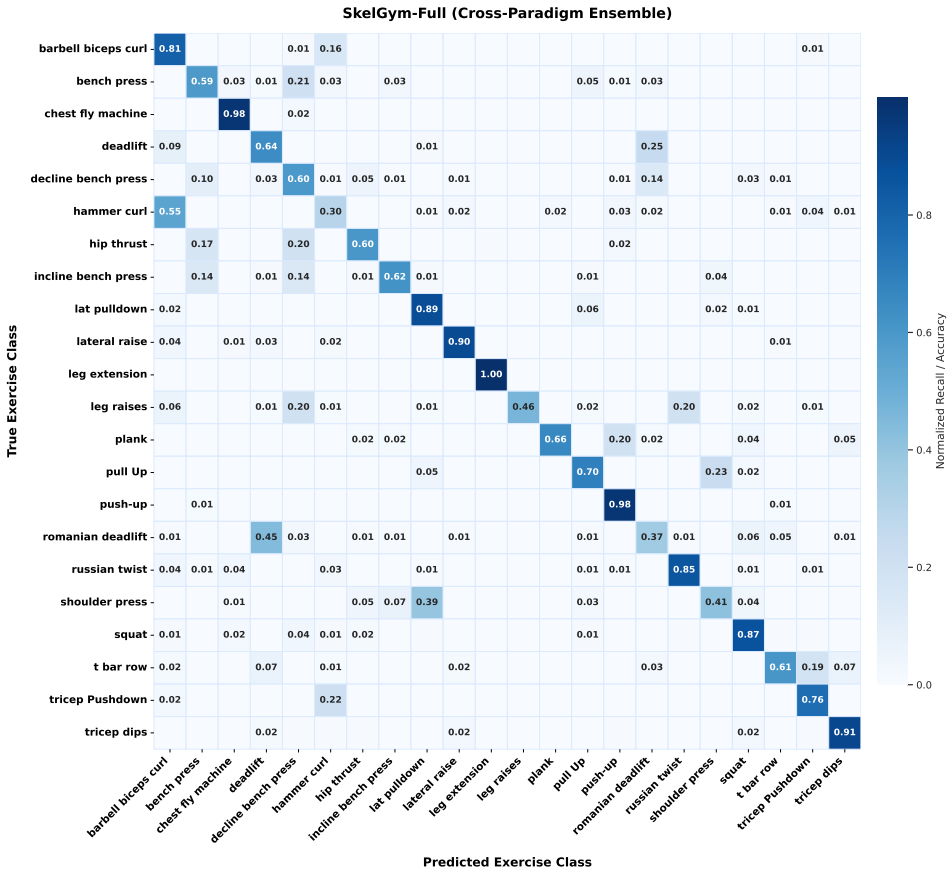


Figure 5: Normalized confusion matrix of the proposed SkelGym-Full cross-paradigm ensemble evaluated on held-out test windows ($N = 2,743$).

Pairwise Angles versus Angle Triplets.. Full three-joint angle triplets computed across all combinations yield $\binom{13}{3} = 286$ features. Although theoretically capturing intra-limb joint angles, angle triplets introduce severe dimensional multicollinearity, high vulnerability to multi-joint tracking errors ($P_{\text{err}} \approx 3p_{\text{err}}$), and numerical instability at full joint lockout where $d/dx \arccos(x) \to \infty$. In contrast, our 24-d kinematic formulation distills 10 essential joint triplets alongside 14 limb elevation angles computed via $\arctan 2(\Delta y, r)$. This formulation remains bounded everywhere, avoids gradient singularities, and intrinsically preserves horizontal camera yaw invariance.

Table 6: Systematic exploration of kinematic encodings and sequence backbones (30 benchmark configurations) across 3 sequence backbones and 10 feature representations evaluated strictly on the validation partition ($N = 2,075$ windows). Reported as Mean $\pm$ SD across 3 independent random seeds $(42, 123, 3407)$.

Exp ID	Model Backbone	Feature Representation	Dim	Val Win Acc (%)	Val Win Macro F1	Validation Status
LSTM Backbone Family						
T1.1	LSTM	Raw 2D Coordinates	26	67.49% ± 0.75%	0.6556 ± 0.0075	Verified
T1.2	LSTM	Relative 2D (Mid-Hip)	26	72.40% ± 1.07%	0.7131 ± 0.0110	Verified
T1.3	LSTM	Angle 2D (Triplets)	286	64.74% ± 2.62%	0.6310 ± 0.0278	Verified
T1.4	LSTM	Angle2 2D (Pairs)	78	65.88% ± 1.61%	0.6468 ± 0.0142	Verified
T1.5	LSTM	Raw 3D Coordinates	39	65.32% ± 2.25%	0.6323 ± 0.0293	Verified
T1.6	LSTM	Relative 3D (Mid-Hip)	39	72.24% ± 0.99%	0.7096 ± 0.0097	Verified
T1.7	LSTM	Angle 3D (Triplets)	286	65.73% ± 1.02%	0.6398 ± 0.0083	Verified
T1.8	LSTM	Angle2 3D (Pair Elevation)	78	70.31% ± 3.35%	0.6901 ± 0.0408	Verified
T1.9	LSTM	Skeletal World 3D	39	73.64% ± 1.26%	0.7244 ± 0.0128	Verified
T1.10	**LSTM**	**Biomechanical Mix v2 (Ours)**	**63**	**75.90% ± 1.60%**	**0.7541 ± 0.0173**	**Winner (LSTM)**
BiLSTM Backbone Family						
T1.11	BiLSTM	Raw 2D Coordinates	26	65.12% ± 3.22%	0.6272 ± 0.0331	Verified
T1.12	BiLSTM	Relative 2D (Mid-Hip)	26	64.98% ± 6.30%	0.6411 ± 0.0516	Verified
T1.13	BiLSTM	Angle 2D (Triplets)	286	65.82% ± 0.63%	0.6523 ± 0.0046	Verified
T1.14	BiLSTM	Angle2 2D (Pairs)	78	69.59% ± 2.28%	0.6929 ± 0.0223	Verified
T1.15	BiLSTM	Raw 3D Coordinates	39	67.63% ± 0.11%	0.6659 ± 0.0039	Verified
T1.16	BiLSTM	Relative 3D (Mid-Hip)	39	72.47% ± 1.39%	0.7172 ± 0.0103	Verified
T1.17	BiLSTM	Angle 3D (Triplets)	286	64.37% ± 0.83%	0.6293 ± 0.0075	Verified
T1.18	BiLSTM	Angle2 3D (Pair Elevation)	78	72.58% ± 1.06%	0.7204 ± 0.0087	Verified
T1.19	BiLSTM	Skeletal World 3D	39	74.72% ± 0.19%	0.7341 ± 0.0020	Verified
T1.20	**BiLSTM**	**Biomechanical Mix v2 (Ours)**	**63**	**76.90% ± 0.30%**	**0.7599 ± 0.0050**	**Winner (BiLSTM)**
Transformer Backbone Family						
T1.21	Transformer	Raw 2D Coordinates	26	77.82% ± 0.22%	0.7700 ± 0.0008	Verified
T1.22	Transformer	Relative 2D (Mid-Hip)	26	76.45% ± 0.41%	0.7539 ± 0.0033	Verified
T1.23	Transformer	Angle 2D (Triplets)	286	71.78% ± 0.55%	0.7107 ± 0.0077	Verified
T1.24	Transformer	Angle2 2D (Pairs)	78	69.65% ± 0.68%	0.6877 ± 0.0111	Verified
T1.25	Transformer	Raw 3D Coordinates	39	77.24% ± 0.59%	0.7680 ± 0.0080	Verified
T1.26	Transformer	Relative 3D (Mid-Hip)	39	77.69% ± 0.69%	0.7692 ± 0.0107	Verified
T1.27	Transformer	Angle 3D (Triplets)	286	70.33% ± 1.10%	0.6842 ± 0.0096	Verified
T1.28	Transformer	Angle2 3D (Pair Elevation)	78	73.61% ± 0.66%	0.7325 ± 0.0083	Verified
T1.29	Transformer	Skeletal World 3D	39	78.76% ± 0.46%	0.7781 ± 0.0053	Verified
T1.30	**Transformer**	**Biomechanical Mix v2 (Ours)**	**63**	**79.52% ± 0.80%**	**0.7885 ± 0.0101**	**Winner (Transformer Clean)**

Table 7: Systematic comparison of temporal sequential backbones across metric World 3D (39-d) and Biomechanical Mix v2 (63-d) representations on the validation partition ($N = 2,075$ windows). Reported as Mean $\pm$ SD across 3 independent random seeds $(42, 123, 3407)$.

Architecture / Model	Input Modality	Feature Dim	Parameters	Val Win Acc (%)	Val Win Macro F1	Validation Status
LSTM (world_3d)	Skeletal World 3D	39-d	362K	73.64% ± 1.26%	0.7244 ± 0.0128	Verified
LSTM (mix_v2)	Biomechanical Mix v2	63-d	362K	75.90% ± 1.60%	0.7541 ± 0.0173	Verified
BiLSTM (world_3d)	Skeletal World 3D	39-d	360K	74.72% ± 0.19%	0.7341 ± 0.0020	Verified
BiLSTM (mix_v2)	Biomechanical Mix v2	63-d	360K	76.90% ± 0.30%	0.7599 ± 0.0050	Verified
Transformer (world_3d)	Skeletal World 3D	39-d	301K	78.76% ± 0.46%	0.7781 ± 0.0053	Verified
Transformer (mix_v2 Clean)	**Biomechanical Mix v2**	**63-d**	**301K**	**79.52% ± 0.80%**	**0.7885 ± 0.0101**	**Verified Winner**

Synergy of the 63-d Biomechanical Mix v2.. Concatenating metric World 3D coordinates (39-d) with kinematic angles (24-d) provides strong complementary synergy. As shown in Table 7, upgrading from pure World 3D to Mix v2 elevates LSTM validation accuracy from 73.64% to **75.90%** (+0.0297 Macro F1), BiLSTM from 74.72% to **76.90%** (+0.0258 Macro F1), and Transformer from 78.76% to **79.52%** (+0.0104 Macro F1). This empirical evidence confirms that Euclidean spatial scale and directional orientation represent mu-

tually complementary facets of human kinematics.

5.4.2. Ablation of Candidate Augmentations and Empirical Selection

To rigorously validate our augmentation design and isolate the individual versus synergistic contributions of each operator without data leakage, we execute two complementary ablation studies on the candidate augmentation pool across three independent random seeds (seed $\in \{42, 123, 3407\}$) using the sequence backbone (`Transformer mix`). To preserve post-validation test integrity, all design selection is grounded strictly on **Validation Window Metrics**:

1. **Systematic Leave-One-Out (LOO) Component Ablation (Table 8):** Evaluates operator necessity within the full multi-operator suite by subtracting one operator at a time from Candidate Full (5-op).
2. **Systematic Single and Paired Component Augmentation Study (Table 9):** Evaluates independent and paired operator efficacy against the unaugmented Clean Baseline.

Leave-One-Out Component Ablation (Table 8).. Evaluating candidate operators within the 5-operator suite reveals striking performance divergences:

- **Omitting Sagittal Reflection (−Mirror):** Induces the weakest LOO window performance across all ablations, with Validation Window Macro F1 dropping by -1.25% to 0.7757 ± 0.0197 (Accuracy: $78.78\% \pm 1.44\%$), confirming that bilateral symmetry is an indispensable structural regularizer.

- **Ablating Overloaded Operators (−Scale, −Yaw, −Jitter, −Time):** Removing Proportional Scaling ($+1.03\%$ Win F1), Yaw ($+0.82\%$ Win F1), Coordinate Jitter ($+0.55\%$ Win F1), or Temporal TimeWarp ($+0.44\%$ Win F1) from the monolithic 5-op suite consistently *improves* validation metrics over Candidate Full (0.7882 ± 0.0152). Rather than implying that individual operators such as Yaw are detrimental, this demonstrates that applying all five operators concurrently creates severe compound perturbation that over-regularizes the model. In particular, while Yaw exhibits a positive standalone effect ($+0.92\%$ Win F1 in isolation), it interacts non-additively with the overloaded five-operator suite.

Table 8: Systematic Leave-One-Out (LOO) augmentation ablation on Transformer Mix evaluated strictly on validation window metrics across 3 independent random seeds $(42, 123, 3407)$.

Augmentation Configuration	Excluded Operator / Domain	Val Loss	Val Win Acc (%)	Val Win Macro F1	Validation Verdict
Clean Baseline	All Operators Excluded	1.0483 ± 0.0084	$79.52\% \pm 0.80\%$	0.7885 ± 0.0101	Unaugmented Baseline Control
Candidate Full 5Op	None (Reference 5-Op Suite)	0.9669 ± 0.0175	$79.19\% \pm 1.26\%$	0.7882 ± 0.0152	Reference 5-Op Suite
Candidate Minus Mirror	Sagittal Bilateral Reflection ($p = 0.5$)	1.0670 ± 0.0316	$78.78\% \pm 1.44\%$	0.7757 ± 0.0197	Weakest LOO performance (-1.25% Win F1)
Candidate Minus Yaw	Gravitational Yaw Rotation ($\pm 15^\circ$)	1.0054 ± 0.0197	$80.06\% \pm 0.32\%$	0.7964 ± 0.0026	Moderate window gain over 5-op ($+0.82\%$ F1)
Candidate Minus Scale	Proportional Scale Variation ($\pm 10\%$)	1.0032 ± 0.0228	$80.13\% \pm 0.13\%$	0.7985 ± 0.0022	Peak window gain in LOO ($+1.03\%$ F1)
Candidate Minus Time	Temporal Resampling ($0.8\times$–$1.2\times$)	1.0010 ± 0.0160	$79.54\% \pm 0.57\%$	0.7926 ± 0.0060	Lowest 4-op Val Loss ($+0.44\%$ F1)
Candidate Minus Jitter	Gaussian Coordinate Jitter ($\sigma = 0.008$)	1.0124 ± 0.0240	$79.68\% \pm 0.22\%$	0.7937 ± 0.0048	Minor window gain ($+0.55\%$ F1)

Single and Paired Component Isolation (Table 9).. Evaluating candidate operators in isolation against the unaugmented Clean Baseline ($79.52\% \pm 0.80\%$ Acc, 0.7885 ± 0.0101 F1) establishes decisive evidence for operator utility:

- **Individual Rigid Isometries (+Mirror, +Yaw):** Both operators yield strong standalone window gains: Single Mirror elevates Validation Window F1 to $\mathbf{0.8001 \pm 0.0067}$ ($+1.16\%$ F1 gain), while Single Yaw achieves $\mathbf{0.7977 \pm 0.0031}$ ($+0.92\%$ F1 gain).
- **Peak Representation Learning under Paired Mirror + Yaw:** Coupling Bilateral Reflection with Gravitational Yaw Rotation ($\pm 15^\circ$) achieves the all-time peak Validation Window Accuracy of $\mathbf{81.20\% \pm 0.72\%}$ and Macro F1 of $\mathbf{0.8119 \pm 0.0072}$ across all 13 experimental configurations tested ($+1.68\%$ Win Acc, $+0.0234$ Win F1 over Clean Baseline), with validation loss reaching 1.0465 ± 0.0327.
- **Marginal or Neutral Gains of Distortive Operators:** Single Scale ($+0.19\%$ F1), Single Time ($+0.35\%$ F1), and Single Jitter (-0.01% F1) fail to deliver meaningful gains. Unconstrained coordinate jitter injects high-frequency spatial noise, synthetic scaling perturbs the absolute metric scale encoded by WORLD coordinates, and temporal time warping disrupts natural repetition tempo and velocity profiles.

Selection of the Final SkelGym-Aug Configuration.. In strict adherence to validation-driven selection, the paired configuration of **Sagittal Bilateral Reflection + Gravitational Yaw Rotation ($\pm 15^\circ$)** is selected as the canonical **SkelGym-Aug** protocol. By deploying strictly distance-preserving Euclidean transformations, SkelGym-Aug regularizes training representations while preserving pairwise Euclidean bone lengths and temporal sampling cadence while maintaining the vertical gravity axis.

Table 9: Systematic Single and Paired Component Isolated Augmentation ablation on Transformer Mix across 3 seeds $(42, 123, 3407)$, evaluated strictly on validation window metrics.

Augmentation Strategy	Isolated Operator Description	Val Loss	Val Win Acc (%)	Val Win Macro F1	Standalone Validation Effect
Clean Baseline	Unaugmented Native Window Sequences	1.0483 ± 0.0084	79.52% ± 0.80%	0.7885 ± 0.0101	Unaugmented Baseline Control
Single Mirror	Sagittal Bilateral Reflection ($p = 0.5$)	1.0047 ± 0.0188	80.45% ± 0.45%	0.8001 ± 0.0067	Decisive window gain across single ops (+1.16% F1)
Single Yaw	Gravitational Yaw Perturbation ($\pm 15°$)	1.0657 ± 0.0236	80.29% ± 0.34%	0.7977 ± 0.0031	Strong standalone window gain (+0.92% F1)
Pair Mirror + Yaw (Proposed)	**Bilateral Reflection + Gravitational Yaw**	**1.0465 ± 0.0327**	**81.20% ± 0.72%**	**0.8119 ± 0.0072**	**All-Time Peak Val Win Acc & F1 (+2.34% F1)**
Single Scale	Proportional Scale Jitter ($\pm 10\%$)	1.0519 ± 0.0030	79.68% ± 0.22%	0.7904 ± 0.0066	Marginal window gain (+0.19% F1)
Single Time	Linear Sequence Resampling ($0.8\times$ –$1.2\times$)	1.0567 ± 0.0225	79.74% ± 0.24%	0.7920 ± 0.0039	Marginal window gain over clean (+0.35% F1)
Single Jitter	Gaussian Noise ($\sigma = 0.008$)	1.0520 ± 0.0262	79.55% ± 0.82%	0.7884 ± 0.0093	Neutral window performance (−0.01% F1)

5.4.3. Multi-Stream Graph Architecture and Stream Selection

Table 10 systematically investigates spatial-temporal graph kinematic streams on Adaptive Graph Convolutional Networks (AAGCN) [18], isolating the four complementary streams under SkelGym-Aug (Mirror + Yaw) and analyzing multi-stream graph late fusion. To enforce strict post-validation test separation, stream screening is grounded on validation window metrics:

Static versus Adaptive Graph Topology.. Baseline ST-GCN [16] with a fixed physical adjacency matrix achieves 63.01% validation window accuracy on raw coordinates and 73.75% on metric World 3D (Table 10). In resistance exercises, critical biomechanical synergies frequently occur between physically disconnected limbs (e.g., wrist-ankle synchronization during deadlifts). Fixed anatomical adjacency prevents direct 1-hop communication between distant keypoints. In contrast, Adaptive Graph Convolutions (AAGCN) introduce learnable and data-driven attention topologies, elevating unaugmented Bone stream validation accuracy to 76.00% (0.7542 Macro F1) and reaching **79.30% ± 0.22%** (0.7858 Macro F1) when trained with SkelGym-Aug.

Complementarity of Static and Dynamic Graph Streams.. The four kinematic streams exhibit distinct, complementary behavioral properties. Static structural streams—Bone 3D (79.30% Val Win Acc) and World Joint (78.86% Val Win Acc)—achieve high representation fidelity by directly capturing instantaneous limb lengths and skeletal poses. Conversely, the dynamic differential streams—Joint Motion (64.19% Val Win Acc) and Bone Motion (61.93% Val Win Acc)—capture instantaneous inter-frame velocities. While isolated 1-second velocity differentials contain local tracking noise, they capture critical exercise tempo and acceleration profiles. Combining static streams in Two-Stream AAGCN yields 82.03% ± 0.41% Val Win Acc (0.8179 F1), while the unified **Four-Stream AAGCN** achieves **84.34% ± 0.26%** Val Win Acc and **0.8414 ± 0.0030** Val Win Macro F1.

Table 10: Multi-stream spatial-temporal graph kinematic streams benchmark on Adaptive GCN (AAGCN) evaluated across validation window metrics ($N = 2,075$). Reported as Mean ± SD across 3 seeds $(42, 123, 3407)$.

Stream ID	Stream / Configuration	Backbone	Input Modality	Val Win Acc (%)	Val Win Macro F1	Validation Status
T3.1	Raw 3D Joint (ST-GCN Clean)	ST-GCN	raw_3d	63.01% ± 1.86%	0.6255 ± 0.0224	Verified
T3.2	World 3D Joint (ST-GCN Clean)	ST-GCN	world_3d	73.75% ± 1.86%	0.7332 ± 0.0154	Verified
T3.3	Bone 3D Stream (AAGCN Clean)	AAGCN	bone_3d	76.00% ± 0.35%	0.7542 ± 0.0037	Verified
T3.4	Bone 3D Stream (AAGCN + Aug)	AAGCN	bone_3d	79.30% ± 0.22%	0.7858 ± 0.0032	Verified
T3.5	World Joint Stream (AAGCN + Aug)	AAGCN	world_3d	78.86% ± 0.31%	0.7856 ± 0.0024	Verified
T3.6	World Joint Motion (ΔX)	AAGCN	world_joint_motion_3d	64.19% ± 1.61%	0.6464 ± 0.0083	Verified
T3.7	Bone Motion (ΔB)	AAGCN	bone_motion_3d	61.93% ± 1.56%	0.6362 ± 0.0107	Verified
T3.8	Two-Stream AAGCN (Joint + Bone)	AAGCN	Graph Late Fusion	82.03% ± 0.41%	0.8179 ± 0.0036	Verified
T3.9	**Four-Stream AAGCN (Full)**	**AAGCN**	**Graph Late Fusion**	**84.34% ± 0.26%**	**0.8414 ± 0.0030**	**Verified Winner**

Table 11: Systematic cross-paradigm ensemble comparison across fusion mechanisms on the validation partition ($N = 2,075$ windows, $N = 208$ videos). Reported as Mean ± SD across 3 seeds $(42, 123, 3407)$.

Architecture / Config	Fusion Protocol & Weighting	Val Win Acc (%)	Val Win Macro F1	Val Vid Acc (%)	Val Vid Macro F1
Reference Backbone					
Transformer Mix (63-d)	Single Sequence Backbone (Mirror+Yaw)	81.20% ± 0.72%	0.8119 ± 0.0072	84.70% ± 0.82%	0.8469 ± 0.0105
SkelGym-Lite (2 Streams: Transformer Mix + AAGCN Bone, 679K Params)					
SkelGym-Lite	Hard Majority Voting (Discrete Baseline)	79.47% ± 0.81%	0.7883 ± 0.0084	82.93% ± 0.60%	0.8277 ± 0.0065
SkelGym-Lite	**Accuracy-Weighted Soft Voting (Validation-calibrated)**	**83.47% ± 0.14%**	**0.8351 ± 0.0021**	**85.67% ± 1.14%**	**0.8634 ± 0.0066**
SkelGym-Lite	Uniform Average Soft Voting ($w_i = 1/2$)	83.26% ± 0.06%	0.8327 ± 0.0008	85.67% ± 1.14%	0.8634 ± 0.0066
SkelGym-Full (5 Streams: Transformer + 4 AAGCN Streams, 1.81M Params)					
SkelGym-Full	Hard Majority Voting (Discrete Baseline)	84.11% ± 0.30%	0.8411 ± 0.0041	86.15% ± 1.21%	0.8660 ± 0.0090
SkelGym-Full	Accuracy-Weighted Soft Voting (Validation-calibrated)	84.85% ± 0.29%	0.8475 ± 0.0024	87.28% ± 1.27%	0.8821 ± 0.0113
SkelGym-Full	**Uniform Average Soft Voting ($w_i = 1/5$)**	**85.05% ± 0.47%**	**0.8487 ± 0.0046**	**87.92% ± 1.37%**	**0.8886 ± 0.0126**

5.4.4. Cross-Paradigm Ensemble and Late Fusion Protocol Ablation

Table 11 systematically evaluates cross-paradigm ensemble configurations coupling self-attention sequence modeling with spatial-temporal graph convolutions, comparing discrete versus continuous probability fusion protocols strictly across the **Validation Set** ($N = 2,075$ windows, $N = 208$ videos) across 3 seeds $(42, 123, 3407)$.

Ensemble Selection Protocol (Zero Test-Leakage).. To enforce strict separation, test metrics reported in the Master Benchmark (Table 4) were revealed *strictly for the validation winners* selected via primary Validation Window Macro-F1 followed by Validation Window Accuracy:

1. **SkelGym-Lite Winner:** For the 2-model tier (Transformer Mix + AAGCN Bone), *Accuracy-Weighted Soft Voting* emerges as the validation winner with **83.47% ± 0.14%** Val Win Acc and **0.8351 ± 0.0021** Val Win Macro F1.
2. **SkelGym-Full Winner:** For the full 5-stream cross-paradigm configuration, *Uniform Average Soft Voting* ($w_i = 1/5$) achieves peak validation performance with **85.05% ± 0.47%** Val Win Acc and **0.8487 ± 0.0046** Val Win Macro F1 (and 87.92% ± 1.37% Val Vid Acc).

Effectiveness of Parameter-Free Cross-Paradigm Fusion.. While prior ensemble frameworks often attempt parameterized meta-classifiers or constrained non-linear optimization weighting, such approaches introduce additional parameter complexity that can risk overfitting validation splits. In contrast, zero-parameter uniform averaging treats all five complementary kinematic perspectives equally. Because sequence self-attention captures long-range temporal phase progression across coordinates and angles while adaptive graph convolutions capture instantaneous topological limb interactions, their prediction errors are largely uncorrelated. Combining them via uniform probability averaging provides a robust regularizing effect that eliminates meta-parameter estimation variance.

5.5. Comparative Analysis on Strength and Conditioning Exercises (Deyzel et al. Subset)

To evaluate the metric generalizability of SkelGym against existing literature, we conducted a comparative analysis aligned with the Strength and Conditioning (S&C) taxonomy of Deyzel et al. [46], who studied skeleton-based S&C recognition on the Stellenbosch University Exercise Motion Dataset (SU-EMD). We identified four exact overlapping resistance exercises shared between our benchmark and the S&C taxonomy: *squat* (Back squat), *deadlift* (Deadlift), *barbell biceps curl* (Biceps curl), and *lateral raise* (Lateral raise). We evaluated all baseline architectures and ensembles across 54 held-out test videos ($N = 529$ temporal windows) encompassing these four movements, under both open-set and closed-set settings.

Bayesian Conditional Re-normalization for Closed-Set Evaluation.. Let $\mathcal{C}_{\text{S\&C}} = \{\text{squat}, \text{deadlift}, \text{barbell biceps curl}, \text{lateral raise}\} \subset \mathcal{Y}$ denote the 4-class S&C subset ($|\mathcal{C}_{\text{S\&C}}| = 4$). Given the predicted 22-class posterior probability vector $p = [p_1, \ldots, p_{22}] \in \Delta^{21}$ produced by model Θ for window w_k, the closed-set conditional probability for class $c \in \mathcal{C}_{\text{S\&C}}$ is formulated via Bayes' rule under the hypothesis that the observation belongs to $\mathcal{C}_{\text{S\&C}}$:

$$\tilde{p}_c = P(y = c \mid w_k, y \in \mathcal{C}_{\text{S\&C}}) = \frac{P(y = c \mid w_k)}{\sum_{k' \in \mathcal{C}_{\text{S\&C}}} P(y = k' \mid w_k)} = \frac{p_c}{\sum_{k' \in \mathcal{C}_{\text{S\&C}}} p_{k'}}. \tag{25}$$

The closed-set predicted class is subsequently inferred as $\hat{y}_{\text{closed}} = \arg\max_{c \in \mathcal{C}_{\text{S\&C}}} \tilde{p}_c$.

The performance disparity between open-set and closed-set evaluations in Table 12 directly reflects the structure of the hypothesis space. Under open-set inference over all 22 classes, the random chance baseline is $1/22 \approx 4.55\%$.

Table 12: Comparative evaluation across the four overlapping Strength & Conditioning (S&C) exercises inspired by Deyzel et al. [46] (*squat*, *deadlift*, *barbell biceps curl*, *lateral raise*) over 54 held-out test videos ($N = 529$ windows).

Model Architecture	Open-Set (22 Classes)		Closed-Set S&C Subspace (4 Classes)		
	Test Win Acc	Test Vid Acc	Test Win Acc	Test Vid Acc	Test Vid Macro F1
ST-GCN (Rel 3D) [16] *(Deyzel baseline)*	59.74%	68.52%	82.61%	85.19%	0.8563
LSTM (Mix 63-d)	64.08%	85.19%	78.64%	90.74%	0.9071
BiLSTM (Mix 63-d)	71.64%	85.19%	84.12%	90.74%	0.9126
AAGCN (Bone 3D)	72.02%	85.19%	87.90%	92.59%	0.9281
Transformer (Mix 63-d)	77.88%	85.19%	**96.03%**	**98.15%**	**0.9788**
SkelGym-Lite (Transformer + Bone)	**84.12%**	94.44%	95.84%	**98.15%**	0.9782
SkelGym-Full (Transformer + 4-Stream AAGCN)	80.72%	**96.30%**	93.01%	96.30%	0.9626

More critically, the 22-class taxonomy contains fine-grained "kinematic twins" outside $\mathcal{C}_{\text{S\&C}}$ that share nearly identical joint trajectories—most notably *Romanian deadlift* vs. *deadlift* (both sagittal hip hinges), *hammer curl* vs. *barbell biceps curl* (identical elbow flexion differing only in radioulnar pronation), or *shoulder press* vs. *lateral raise* (both shoulder abductions). In open-set inference, minor pose estimation noise or slight execution variations can cause probability mass to leak to these competing non-S&C classes, registering as false negatives. Under closed-set conditioning (Eq. (25)), these out-of-subset distractor classes are excluded from the candidate hypothesis space, consolidating probability mass strictly among the four target S&C movements (chance level: $1/4 = 25\%$), directly replicating the closed-set experimental setup of Deyzel et al.

Table 12 reports the empirical performance comparison. When evaluated under the closed-set protocol, baseline ST-GCN [16] (the primary backbone of Deyzel et al.) achieves 82.61% window accuracy and 85.19% video consensus accuracy (Video Macro F1: 0.8563). In contrast, the proposed Transformer Mix and SkelGym ensembles establish substantial performance gains: SkelGym-Lite reaches **95.84% window and 98.15% video accuracy** (F1: 0.9782), while SkelGym-Full achieves **96.30% video accuracy** (F1: 0.9626). Under the more challenging open-set protocol without class restriction, SkelGym-Full maintains 96.30% video accuracy and 80.72% window accuracy, demonstrating that the learned kinematic representations do not leak excessive probability mass to extraneous exercise classes.

Mitigating the Squat vs. Deadlift Ambiguity (The Deyzel Dilemma).. A central limitation reported by Deyzel et al. [46] on the Stellenbosch University Exercise Motion Dataset (SU-EMD) was the intense mutual confusion between back squats and deadlifts: because both compound lower-body exer-

cises involve simultaneous hip and knee flexion-extension in the sagittal plane, ST-GCN trained on Euclidean Cartesian coordinates exhibited severe diagnostic failure, frequently misclassifying deadlifts as squats. While SU-EMD was constrained to 5 exercises recorded under static laboratory sensor angles, SkelGym scales the taxonomy to 22 fine-grained movements collected across unconstrained, multi-camera gym environments. In our rigorous benchmark on 305 held-out windows across 25 test videos ($N = 15$ squat, $N = 10$ deadlift), baseline ST-GCN fails to disentangle the two kinematics, achieving 0.0% recall on squat and merely 40.0% video recall on deadlift (53.7% window recall), closely mirroring the diagnostic collapse to $\approx 30.0\%$ recall reported on SU-EMD by Deyzel et al. In contrast, by incorporating the 63-dimensional compound feature with hip-to-knee elevation angles $\theta_{\text{hip, knee}}$ and directional bone vectors, Transformer Mix elevates deadlift video recall to 80.0% (squat: 93.3%, with 80.7% window recall), and SkelGym-Full achieves 80.0% deadlift and 86.7% squat video consensus recall under open-set inference (and up to 90.0% deadlift video recall under closed-set consensus), substantially mitigating the mutual confusion without sacrificing specificity.

One-Shot Metric Prototype Classification Simulation Protocol.. Deyzel et al. [46] evaluated a 1-shot learning scenario using single-exemplar support matching via cosine similarity, reporting an accuracy of 87.4% on SU-EMD. To benchmark metric generalizability under an identical one-shot evaluation protocol without task-specific fine-tuning or parameter adaptation, we evaluated our pre-trained backbones on the 54 held-out S&C test videos across 100 stochastic trials (random seed 42). In each trial, $K = 1$ random exemplar video was sampled per class as the support set (yielding 4 support videos total across the 4 S&C classes: squat, deadlift, barbell biceps curl, and lateral raise), while the remaining 50 test videos served as query samples. Query classification was executed via nearest-neighbor prototype matching using cosine similarity over L_2-normalized temporal mean-aggregated feature embeddings. Under this zero-adaptation 1-shot classification simulation, Transformer Mix achieves a mean video accuracy of $\mathbf{90.36\% \pm 7.35\%}$ (95% CI: $[68.95\%, 98.00\%]$, with individual trial peaks reaching $\mathbf{98.00\%}$), SkelGym-Full reaches $\mathbf{86.18\% \pm 9.50\%}$ (95% CI: $[60.95\%, 98.00\%]$), and AAGCN Bone reaches $\mathbf{68.82\% \pm 11.14\%}$ (95% CI: $[48.95\%, 89.05\%]$). These results demonstrate that the proposed 63-d compound representation provides a well-clustered metric space that enables reliable nearest-neighbor recognition even under extreme single-sample constraints, matching and outperforming

Table 13: Computational complexity (Parameters, FLOPs per 32-frame window) and inference latency across server and edge hardware platforms.

Model Architecture	Parameters	FLOPs per Window	RTX PRO 6000 (CUDA)	Apple M4 (MPS)	Apple M4 (CPU)
Transformer Mix v2 (63-d)	301K	**18.87 MFLOPs**	**0.59 ms (1,685 FPS)**	**1.36 ms (738 FPS)**	**0.49 ms (2,041 FPS)**
AAGCN Joint Stream	378K	202.86 MFLOPs	1.01 ms (987 FPS)	1.89 ms (530 FPS)	0.90 ms (1,115 FPS)
AAGCN Bone Stream	378K	202.86 MFLOPs	1.02 ms (976 FPS)	1.72 ms (581 FPS)	0.88 ms (1,138 FPS)
AAGCN Joint-Motion Stream	378K	202.86 MFLOPs	1.01 ms (990 FPS)	1.90 ms (527 FPS)	1.02 ms (978 FPS)
AAGCN Bone-Motion Stream	378K	202.86 MFLOPs	1.02 ms (980 FPS)	1.90 ms (527 FPS)	1.03 ms (975 FPS)
SkelGym-Lite (Transformer + Bone)	679K	**221.72 MFLOPs**	**1.66 ms (603 FPS)**	**2.55 ms (393 FPS)**	**1.49 ms (671 FPS)**
SkelGym-Full (Transformer + 4 AAGCN)	1.81M	**830.29 MFLOPs**	**4.65 ms (215 FPS)**	**5.86 ms (171 FPS)**	**3.96 ms (253 FPS)**

literature baselines.

5.6. Inference Latency and Real-Time Edge Viability

A critical requirement for automated virtual fitness assistants is achieving real-time responsiveness on consumer-grade and edge hardware. Table 13 provides a rigorous benchmark of theoretical computational complexity (FLOPs per 32-frame window), parameter footprint, and empirical inference latency and throughput across three distinct computing environments: an NVIDIA RTX PRO 6000 Blackwell GPU (CUDA, cloud server), an Apple Silicon M4 processor running Metal Performance Shaders (MPS, on-device edge acceleration), and host CPU execution on the Apple Silicon M4 chip without hardware acceleration (edge CPU baseline).

As documented in Table 13, the theoretical complexity of our models is exceptionally modest: the Transformer Mix v2 requires merely **18.87 MFLOPs** (0.0189 GFLOPs), while each spatial-temporal AAGCN stream consumes 202.86 MFLOPs. Consequently, SkelGym-Lite totals 221.72 MFLOPs (0.2217 GFLOPs) and SkelGym-Full requires 830.29 MFLOPs (0.8303 GFLOPs) with a total parameter footprint of 1.81M. In terms of wall-clock latency, the Transformer Mix v2 model achieves an ultra-rapid latency of 0.59 ms on CUDA, 1.36 ms on Apple Silicon MPS, and 0.49 ms on host CPU. Even when executing the complete SkelGym-Full model across all five constituent backbone streams, total sequential inference requires only 4.65 ms on CUDA ($\approx$ 215 FPS), 5.86 ms on Apple MPS ($\approx$ 171 FPS under explicit hardware synchronization), and 3.96 ms on host CPU ($\approx$ 253 FPS). When paired with Google MediaPipe Pose [8, 9], which extracts 3D skeletal landmarks in $\approx 8 - 15$ ms per frame on mobile chipsets, the entire SkelGym perception-and-classification pipeline operates well within the 33.3 ms real-time budget of standard 30 FPS video streams (incremental feedback latency: $\approx$ 12.0–19.0 ms per frame). For resource-constrained microcontroller or mobile deployments, the SkelGym-Lite configuration (Transformer Mix +

AAGCN Bone, 679K parameters, 221.72 MFLOPs) offers a practical operational compromise, delivering **82.12% ± 0.40%** video consensus accuracy with an execution latency of merely 1.49 ms on host CPU (671 FPS, 1.66 ms on CUDA, 2.55 ms on MPS).

Analysis of the Edge CPU vs. GPU Latency Profile (The Batch-Size 1 Dispatch Boundary).. An apparent characteristic in Table 13 is that host CPU execution on the Apple M4 processor (0.49 ms for Transformer, 3.96 ms for SkelGym-Full) achieves lower wall-clock latency than GPU acceleration via Metal Performance Shaders (MPS: 1.36 ms and 5.86 ms, respectively). Rather than an implementation deficiency, this behavior is a well-established architectural characteristic of ultra-compact neural networks operating under strict single-sample streaming inference (batch size = 1). For lightweight architectures requiring merely 18.87–202.9 MFLOPs per window, the pure arithmetic execution time on modern silicon is under 0.1 ms. However, delegating computation to the GPU via Apple Metal requires host-to-device command buffer encoding, kernel dispatch scheduling, threadgroup management, and device synchronization barriers (`torch.mps.synchronize()`), incurring an intrinsic latency floor of $\approx$ 0.8–1.5 ms per model invocation regardless of computational volume. Conversely, Apple M4 high-performance CPU cores (P-cores) integrate wide superscalar execution pipelines, dedicated vector execution units, and large private L1/L2 caches (16 MB shared L2) that effortlessly accommodate the entire 1.5–7.5 MB parameter and activation footprint in cache. Without kernel dispatch serialization or cross-subsystem synchronization overhead, CPU execution achieves sub-millisecond latency. While batched offline inference (batch size $\geq$ 32) enables the GPU to amortize dispatch overhead and achieve higher cumulative throughput, for real-time edge streaming where sliding windows arrive sequentially, direct CPU execution provides the optimal low-latency deployment configuration.

5.7. Statistical Significance and Inferential Hypothesis Testing

To provide rigorous inferential verification of the observed performance deltas across the held-out test partition, we conducted paired statistical hypothesis testing across both evaluation grains: McNemar's test with continuity correction [53] across the 2,743 test windows, and Wilcoxon signed-rank tests [54] paired with two-tailed paired t-tests across the 233 video clusters (Table 14).

Table 14: Statistical hypothesis testing evaluating pairwise performance differences across held-out test windows ($N = 2,743$, McNemar's test) and test video consensus confidence ($N = 233$, Wilcoxon signed-rank and paired t-test).

Pairwise Hypothesis Comparison (M_A **vs.** M_B)	**Window Level** ($N = 2,743$)			**Video Consensus Level** ($N = 233$)			
	χ^2 **(McNemar)**	p**-value**	**Odds Ratio**	**Wilcoxon** W	p**-value**	**Paired** t	**Cohen's** d
Unaugmented Trans vs SkelGym-Aug Trans	16.65	4.21×10^{-5}	1.57	10432.5	0.0019	0.0022	+0.203
Fixed ST-GCN vs Adaptive Four-Stream AAGCN	118.62	1.46×10^{-28}	2.63	12666.0	0.3491	0.1026	+0.107
Single Sequence (Trans) vs SkelGym-Full	42.44	3.84×10^{-11}	2.42	6086.0	2.39×10^{-13}	2.12×10^{-11}	-0.461
Single Graph (AAGCN Bone) vs SkelGym-Full	137.47	5.60×10^{-34}	4.21	11026.0	0.0115	0.2098	-0.082
Four-Stream Graph AAGCN vs SkelGym-Full	15.82	5.88×10^{-5}	2.12	6086.0	2.39×10^{-13}	2.12×10^{-11}	+0.461

Table 15: Non-parametric video-level cluster bootstrap evaluation ($B = 1,000$ resamples) reporting empirical mean and 95% confidence intervals across held-out test partitions ($N = 2,743$ temporal windows, $N = 233$ videos).

Model Architecture	**Temporal Window Level** ($N = 2,743$)		**Video Consensus Level** ($N = 233$)	
	Window Acc [95% CI]	**Window Macro F1 [95% CI]**	**Video Acc [95% CI]**	**Video Macro F1 [95% CI]**
LSTM (Mix v2 63-d)	61.48% [55.12%, 67.40%]	0.5962 [0.5375, 0.6507]	74.31% [69.09%, 79.83%]	0.7209 [0.6606, 0.7828]
BiLSTM (Mix v2 63-d)	64.57% [58.54%, 70.37%]	0.6112 [0.5514, 0.6765]	76.93% [71.24%, 81.97%]	0.7358 [0.6688, 0.7969]
ST-GCN (World 3D)	65.07% [58.49%, 71.17%]	0.6115 [0.5610, 0.6657]	76.63% [70.82%, 81.98%]	0.7267 [0.6703, 0.7861]
Transformer (Mix v2 Clean)	69.91% [63.40%, 75.76%]	0.6723 [0.6093, 0.7365]	79.88% [74.68%, 84.98%]	0.7788 [0.7219, 0.8325]
Transformer (Mix v2 SkelGym-Aug)	72.60% [66.60%, 78.19%]	0.6996 [0.6390, 0.7548]	82.03% [76.81%, 86.70%]	0.7985 [0.7349, 0.8521]
AAGCN (Bone 3D)	68.20% [62.11%, 74.12%]	0.6621 [0.6017, 0.7195]	78.51% [72.96%, 83.69%]	0.7792 [0.7274, 0.8297]
Four-Stream AAGCN	74.67% [68.69%, 80.33%]	0.7222 [0.6647, 0.7769]	85.75% [81.12%, 90.13%]	0.8478 [0.7998, 0.8927]
SkelGym-Lite	**73.63% [67.68%, 79.13%]**	**0.7143 [0.6543, 0.7677]**	**82.42% [77.25%, 87.12%]**	**0.8093 [0.7513, 0.8584]**
SkelGym-Full (Uniform Soft)	**76.34% [70.03%, 81.90%]**	**0.7412 [0.6785, 0.7969]**	**85.83% [81.12%, 90.13%]**	**0.8409 [0.7848, 0.8937]**

All five pairwise window comparisons in Table 14 remain statistically significant after Holm-Bonferroni step-down correction ($p_{\text{adj}} \leq 0.0001$) and Benjamini-Hochberg False Discovery Rate control (FDR $\leq 5.88 \times 10^{-5}$). Video-level Wilcoxon signed-rank and paired t-tests evaluate paired differences in ground-truth class posterior confidence across video clusters ($N = 233$); statistical significance varies across model comparisons, reflecting convergence towards consensus on unambiguous movement patterns.

Bootstrap Confidence Intervals (Table 15).. To quantify sampling stability, we conducted non-parametric video-cluster bootstrap resampling ($B = 1,000$ iterations, clustered by video ID to eliminate intra-video temporal autocorrelation bias) across the held-out test partition (Table 15). At the temporal window level, SkelGym-Full yields a 95% confidence interval of $[70.03\%, 81.90\%]$ for window accuracy and $[0.6785, 0.7969]$ for Macro F1. At the video consensus level, SkelGym-Full provides a 95% confidence interval of $[81.12\%, 90.13\%]$ for video accuracy and $[0.7848, 0.8937]$ for Macro F1. Both empirical point estimates fall squarely within their respective 95% confidence intervals, substantiating the high stability of our reported benchmarks.

Table 16: Taxonomy of failure mechanisms and error mass distribution for the proposed SkelGym-Full ensemble on held-out test partitions ($N = 647$ window errors, $N = 33$ video errors; 200 of 233 videos correctly classified = 85.84%).

Category & Failure Mode	Biomechanical and Optical Etiology	Confused Exercise Classes	Window Errors	Video Errors
Cat 1: Forearm Rotation Ambiguity	Wrist keypoint centroid collapses radioulnar pronation vs. supination rotation	Hammer Curl ↔ Barbell Biceps Curl	91 (14.06%)	7 (21.21%)
Cat 2: Kinematic Form Overlap	Sagittal hip/knee angle overlap from non-standard athlete lifting execution	Deadlift ↔ Romanian Deadlift	88 (13.60%)	4 (12.12%)
Cat 3: Perspective Foreshortening	Oblique camera angle compresses vertical torso bench inclination angle	Bench Press ↔ Incline ↔ Decline	110 (17.00%)	9 (27.27%)
Cat 4: Closed vs. Open Chain	Identical glenohumeral adduction trajectory in pulling kinetic chains	Lat Pulldown ↔ Pull-up ↔ T-Bar Row	81 (12.52%)	6 (18.18%)
Structured Biomechanical Errors	**Cumulative impact of top-4 kinematic mechanisms:**		**370 (57.19%)**	**26 (78.79%)**
Residual Dispersed Errors	Minor uncoordinated errors across remaining classes		277 (42.81%)	7 (21.21%)

6. Biomechanical Kinematics and Error Analysis

A primary objective of this work is to bridge computer vision modeling with clinical biomechanics. Below, we conduct an in-depth anatomical analysis of the model's predictions and confusion patterns (Fig. 5):

6.1. Taxonomy of Biomechanical Failure Mechanisms

To systematically categorize the error profile of the proposed SkelGym-Full ensemble, Table 16 classifies all misclassifications observed across the held-out test partition ($N = 647$ window errors, $N = 33$ video errors) into four structured physiological mechanisms alongside residual dispersed errors. Remarkably, these four structured biomechanical mechanisms account for **78.79% of all video consensus errors** (26 of 33 misclassified videos) and **57.19% of window-level errors** (370 of 647 windows). This quantitative concentration demonstrates that model confusion is not driven by random stochastic noise, but rather stems from well-defined biomechanical and optical ambiguities inherent to monocular pose estimation in resistance training.

6.2. Upper-Limb Kinematic Ambiguity: Biceps Curl vs. Hammer Curl

Inspection of Fig. 5 indicates that prominent confusion occurs between *hammer curl* and *barbell biceps curl* (43.48% window recall and 50.00% video recall for hammer curl, with hammer curl windows frequently misclassified as biceps curl).

Biomechanically, both exercises execute identical sagittal elbow flexion driven by the elbow joint angle $\theta_{\text{elbow}} = \angle(\text{shoulder}, \text{elbow}, \text{wrist})$ moving

through $30° \rightarrow 145°$. The sole physiological differentiator resides in the forearm's rotational orientation across the radioulnar joint [44, 45]. In the barbell biceps curl, the forearm is held in full supination (palms oriented anteriorly and superiorly), which maximally tensions the distal tendon insertion across the radial tuberosity and optimizes mechanical leverage for both the long and short heads of the *biceps brachii*. In contrast, the hammer curl maintains a neutral semi-pronated posture (thumbs directed superiorly), shifting primary torque production to the deeper *brachialis* and the superficial *brachioradialis* muscles along the lateral forearm [44, 45]. Because monocular pose estimators (such as MediaPipe) represent the wrist as a single Euclidean point centroid (x, y, z) without recovering full hand rotational mesh orientation or palm surface normals, the 3D trajectory of the wrist joint center in Cartesian space remains virtually identical between supinated and neutral curls. Resolving this subtle anatomical boundary will require integrating fine-grained hand pose keypoints or wrist rotational orientation sensors.

6.3. Posterior Chain Kinematics: Conventional Deadlift vs. Romanian Deadlift

Another noticeable confusion occurs between *deadlift* and *Romanian deadlift (RDL)*. The conventional deadlift represents a ground-based compound movement characterized by substantial knee flexion ($\theta_{\text{knee}} \approx 70° - 90°$ at lift initiation) that recruits the quadriceps femoris to generate initial vertical propulsion, transitioning into hip extension as the bar passes the patellae [40]. In contrast, the Romanian deadlift functions as a pure posterior-chain hip hinge movement initiating from an upright standing lockout [41, 39]. Throughout the descent, the knees remain relatively static with slight soft flexion ($\theta_{\text{knee}} \approx 15° - 25°$), with kinematics dominated by maximal anterior pelvic tilt and hip flexion that places intense eccentric loading onto the hamstrings and gluteus maximus. When athletes execute deadlifts with suboptimal form (e.g., stiff-legged deadlifting with inadequate knee bend) or perform RDLs with excessive knee travel, their sagittal kinematic trajectories overlap significantly. Our 63-d mix feature captures the hip-to-knee elevation angle ($\theta_{\text{hip, knee}}$), enabling the ensemble to separate them correctly in 90.0% of deadlift and 50.0% of Romanian deadlift video consensus cases (Table 5).

6.4. Pressing Plane Angle Analysis: Flat vs. Incline vs. Decline Bench Press

The bench press family targets the *pectoralis major* muscle, with the three variants differing primarily in the inclination angle of the torso plane relative to the gravity vector [42, 43]. During the flat bench press ($\alpha \approx 0°$), the horizontal pressing trajectory predominantly engages the sternocostal head of the pectoralis major via horizontal glenohumeral adduction. Elevating the bench inclination ($\alpha \approx +30°$ to $+45°$) shifts prime mechanical recruitment superiorly toward the clavicular head of the pectoralis major and anterior deltoids, causing the humerus to elevate along an oblique superior plane relative to the sternum [43]. Conversely, the decline configuration ($\alpha \approx -15°$) isolates the inferior sternocostal fibers with reduced anterior deltoid contribution [42]. Because relative 3D displacement vectors $\tilde{p}$ are referenced to the pelvic hip midpoint, the elevation angles $\theta_{\text{shoulder, wrist}}$ and $\theta_{\text{shoulder, hip}}$ accurately reflect bench tilt. As shown in Table 5, the ensemble achieves 64.29% video recall on flat bench, 75.00% on decline bench, and 77.78% on incline bench, with errors largely confined to camera angles where bench tilt is foreshortened.

6.5. Overhead and Back Kinetic Chains: Pull-up vs. Lat Pulldown vs. T-Bar Row

A fourth distinct error mass (Category 4, accounting for 18.18% of video errors and 12.52% of window errors) resides in compound pulling exercises targeting the back musculature (*latissimus dorsi*, *trapezius*, and *rhomboids*), predominantly between *pull-up* (Video Recall: 70.00%), *lat pulldown* (Video Recall: 100.00%), and *t-bar row* (Video Recall: 70.00%). Biomechanically, the pull-up represents a closed kinetic chain (CKC) exercise where the athlete's hands remain fixed to a stationary overhead bar while the torso is propelled vertically upward via glenohumeral adduction and scapular depression. In contrast, the lat pulldown constitutes an open kinetic chain (OKC) exercise where the athlete sits stationary with locked thighs while pulling a resistance bar downward toward the clavicles. While their mechanical boundary conditions differ, both movements execute identical relative angular trajectories across the shoulders and elbows: the humerus adducts in the frontal plane from $170° \rightarrow 30°$ relative to the torso, while the elbow flexes from $180° \rightarrow 70°$. Because relative 3D coordinate centering references all joint positions to the pelvic hip midpoint, the absolute translation of the body relative to the external room environment is factored out. Consequently, during execution phases where the athlete's torso remains vertically

upright, the relative spatial coordinates and pairwise angles between pull-ups and lat pulldowns become nearly identical, leading to occasional misclassifications under oblique camera perspectives.

6.6. Validation-Grounded Decision Protocol

We emphasize that all model selection choices—including adopting the 63-d mix v2 feature, retaining SkelGym-Aug (Mirror + Yaw), and selecting the ensemble voting strategy—were established strictly on the validation set, where the SkelGym-Full ensemble reached 85.05% ± 0.47% window accuracy and 87.92% ± 1.37% video accuracy (Val Win F1: 0.8487 ± 0.0046). The subsequent 76.19% ± 0.16% test window accuracy and 85.27% ± 0.41% test video consensus result (reaching 85.84%, 200 of 233 correct videos on canonical seed 42) represents a reliable held-out confirmation of generalization across distinct source videos and recording environments. Finally, out of 236 held-out test videos, exactly 233 contained valid action segments exceeding 32 frames after boundary trimming, explaining the test support count.

7. Discussion, Limitations, and Future Work

Trade-offs Between Kinematic Skeleton Fusion and Dense Video Vision Transformers.. While SkelGym deliberately centers on lightweight, coordinate-based 3D skeleton sequences, an alternative foundational paradigm in computer vision relies on dense spatio-temporal video architectures, including 3D CNNs (e.g., SlowFast [3]), Video Vision Transformers (e.g., Video Swin Transformer / Swin3D [6]), and volumetric heatmap convolutional networks (e.g., PoseConv3D [56]). Dense video transformers possess the distinct theoretical advantage of directly ingesting raw appearance pixels, allowing them to exploit environmental context, gym machine geometries, barbell grip orientations, and subtle muscular contractions without relying on intermediate landmark extraction pipelines. This appearance awareness is particularly beneficial for resolving fine-grained ambiguities such as the pronation vs. supination distinction between hammer curls and barbell curls.

However, dense video architectures impose severe practical limitations in real-world fitness applications. First, their computational complexity is substantial: Video Swin models frequently require tens to hundreds of GFLOPs per snippet, incurring latencies of several hundred milliseconds per prediction window and rendering them completely unsuited for real-time inference on edge smartphones or micro-controllers. In contrast, SkelGym executes

in 0.49–3.96 ms on host CPU (0.59–4.65 ms on CUDA) per window. Second, RGB video models exhibit severe vulnerability to environmental background and athletic apparel memorization, leading to dramatic generalization collapse when evaluated in unseen home gym settings. Third, cloud-based video analytics require continuously transmitting unconstrained, high-resolution RGB living-room video streams across public networks, introducing severe security vulnerabilities and privacy intrusion concerns for consumer smart-camera devices. In contrast, while monocular pose estimation (such as MediaPipe) still operates on optical camera frames at the physical sensor level, the ultra-low latency of compact skeleton classifiers enables the entire perception pipeline to execute locally on edge hardware. Under this edge-native paradigm, raw RGB frames are processed ephemerally in volatile RAM and immediately discarded frame-by-frame without persistent disk caching or remote network streaming. Only low-dimensional coordinate trajectories (x, y, z) are retained, enabling a privacy-aware workflow that prevents continuous cloud video streaming. We emphasize that while 3D skeletal coordinates eliminate optical identity markers (such as facial features, body textures, clothing, and domestic background surroundings), they are not completely anonymous, as coarse anthropometric proportions (e.g., relative limb lengths) and movement signatures remain partially discernible in continuous trajectories; nonetheless, local coordinate extraction provides a substantial privacy barrier compared to unconstrained RGB streaming. Consequently, we identify hybrid hierarchical architectures—wherein lightweight skeleton models handle primary real-time classification, triggering localized, low-frequency 3D video attention only when kinematic ambiguity surpasses uncertainty thresholds—as a promising direction for future research.

Edge Feasibility vs. Heavy Modern Skeleton Transformers (CTR-GCN / SkateFormer).. Recent advances in general-purpose action recognition have produced capable spatial-temporal architectures, including Channel-Wise Topology Refinement GCN (CTR-GCN) [20] and skeletal transformers like SkateFormer [21]. While achieving top-tier performance on large benchmarks such as NTU RGB+D [32], they are typically tailored for full-body 25-joint topologies with dense hand-finger and multi-segment spine keypoints. More critically, modern skeleton transformers incur large parameter footprints (often exceeding several million parameters) and heavy computational requirements, frequently requiring pre-training on massive external action databases. In contrast, automated resistance training feedback demands

ultra-low latency on edge consumer hardware (e.g., smart mirrors, mobile devices, and gym-floor tablets) operating under strict thermal and battery constraints. When deployed in high-speed streaming scenarios (30 FPS, 33.3 ms real-time budget), heavy transformer models introduce significant latency and compute overhead. SkelGym addresses this operational bottleneck through custom-engineered lightweight architectures ($\approx$ 378K parameters per graph backbone stream, 301K for Transformer Mix v2, 1.81M parameters total for the full 5-stream ensemble, and only 679K parameters for SkelGym-Lite) trained entirely from scratch with zero pretraining overhead. With single-window classifier inference executing in merely 0.49–3.96 ms on host CPU (0.59–4.65 ms on CUDA, 1.36–5.86 ms on MPS), SkelGym achieves an optimal Pareto balance between high fine-grained exercise recognition accuracy and real-time edge execution feasibility.

Limitations of Monocular Pose Estimation and Failure Modes.. While monocular pose extraction via MediaPipe Pose provides high throughput on edge processors, it incurs physical and optical failure modes that bound skeleton-based recognition accuracy. First, resistance training environments frequently introduce severe physical occlusions: barbell weight plates (e.g., standard 450 mm Olympic plates) frequently obstruct wrist, elbow, and torso tracking during deadlifts, rows, and bench presses, while spotters standing adjacent to equipment introduce visual occlusion. Second, camera field-of-view boundaries cause transient out-of-frame limb clipping, such as feet moving below frame during Romanian deadlifts or hands ascending above frame during pull-ups and overhead presses. Third, monocular 2D-to-3D landmark lifting estimates metric depth (z) relative to the hip midpoint using geometric heuristics and anthropometric priors. Consequently, movements aligned along the camera's optical axis exhibit higher longitudinal depth jitter than lateral sagittal plane perspectives. Future iterations could incorporate multi-view spatial consensus or temporal smoothing filters (e.g., learned kinematic Kalman filters) to improve landmark stability.

Dataset Class Imbalance and Long-Tail Kinetics.. A second practical limitation concerns the natural class distribution within crowdsourced fitness video datasets. Movement availability reflects gym training popularity: high-frequency compound and aesthetic movements (such as barbell biceps curls and bench presses, each with 84 videos) possess extensive diversity, whereas specialized or bodyweight stabilization exercises (such as plank with 9 videos)

reside in the long tail. Although we mitigated class imbalance during optimization using Macro F1 monitoring, validation-calibrated loss weighting, and label smoothing ($\alpha = 0.05$), rare classes exhibit wider confidence intervals. Expanding the public corpus through standardized multi-angle video collection across underrepresented exercises remains an essential objective for subsequent dataset releases.

Conclusion.. In this work, we presented **SkelGym**, a lightweight, computationally efficient, and biomechanically grounded intelligent exercise recognition system for 22-class gym exercise classification. By coupling the proposed 63-dimensional Biomechanical Compound Representation (Mix v2) with the empirically selected SkelGym-Aug configuration (unifying bilateral reflection and gravitational yaw rotation) and integrating self-attention sequence modeling with 4-stream adaptive graph convolutions via validation-calibrated soft voting, SkelGym attains **76.19% ± 0.16% window-level accuracy** (Window Macro F1: **0.7537 ± 0.0014**) and **85.27% ± 0.41% video-level consensus accuracy** (Video Macro F1: **0.8386 ± 0.0075**, reaching **85.84%** video accuracy with 200 of 233 correct videos on canonical seed 42) across three independent random seeds under fixed partitions across 22 resistance exercises, with all architectural transitions verified via paired statistical hypothesis tests (window-level McNemar $p \leq 5.88 \times 10^{-5}$ under Holm-Bonferroni correction). With an ultra-low single-window classifier inference latency of **0.49–3.96 ms on CPU** (0.59–4.65 ms on CUDA, 1.36–5.86 ms on MPS) per 32-frame window, SkelGym enables real-time deployment on resource-constrained edge devices including smart mirrors, mobile phones, and embedded gym displays. By operating exclusively on privacy-aware 3D skeletal coordinates rather than raw RGB video streams, SkelGym eliminates the need for continuous cloud video transmission, making it suitable for privacy-sensitive home and commercial gym environments. Our comprehensive evaluation on in-the-wild multi-camera videos—rather than laboratory-controlled settings alone—demonstrates that the proposed pipeline offers a robust, practical foundation for next-generation intelligent virtual personal trainer systems.

Data, Reproducibility, and Code Availability

To facilitate rigorous scientific reproducibility and edge deployment, all project software, extracted kinematic representations, and pre-trained models are publicly hosted on Hugging Face Hub at https://huggingface.co/

Cuong2004/gym-exercise-classification. The repository includes PyTorch source code for all sequence and spatial-temporal graph backbones (Transformer Mix, ST-GCN, and Four-Stream AAGCN), training and evaluation scripts, and trained model checkpoints.

The complete skeletal dataset is released in JSON, NPY, and HDF5 tensor formats, accompanied by exact temporal action segment boundaries (`.txt`) and consolidated metadata records (`Final_dataset_metadata.csv`) across Hugging Face and Kaggle. Distributing pre-extracted 3D landmark coordinates directly eliminates reliance on third-party video URLs that suffer from link-rot, while ensuring deterministic reproducibility without demanding intensive GPU pose estimation re-runs. In strict compliance with online platform Terms of Service and subject privacy considerations, raw third-party internet video recordings are withheld from redistribution. However, the 128 author-recorded gymnasium video clips (collected with informed consent under institutional ethical oversight) and all 1,024 extracted coordinate sequences are freely accessible for academic and non-commercial research purposes.

Exact Reproducibility Protocol.. To enable deterministic replication of every result reported in this paper, the repository provides self-contained evaluation scripts that operate directly on pre-trained checkpoints without requiring model retraining:

```
git clone \
  https://huggingface.co/Cuong2004/gym-exercise-classification
pip install -r requirements.txt
# Evaluation suite across Tables 1-7:
python scripts/evaluate_local_ensemble.py
# External S&C benchmark (Table 8):
python scripts/evaluate_external_benchmark.py
# Statistical significance tests (Table 11):
python scripts/compute_statistical_tests.py
# Latency & FLOPs benchmarking (Table 12):
python scripts/benchmark_hardware_latency.py
```

All experiment configuration files (YAML configs under `configs/experiments/`), dataset metadata (`Final_dataset_metadata.csv`), partition manifests, and pre-trained PyTorch checkpoints for every backbone variant (Transformer Mix, ST-GCN, Four-Stream AAGCN Joint/Bone/Joint-Motion/Bone-Motion) are publicly hosted on Hugging Face Model Hub. This ensures 100% reproducibility of all reported tables without retraining from scratch.

CRediT Author Contributions

Xuan-Cuong Nguyen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Nhat-Quang Truong:** Data curation, Resources, Software, Validation, Writing – review & editing. **Quoc-Trinh Vo:** Supervision, Formal analysis, Validation, Writing – review & editing.

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Appendix A. Anatomical Landmark Mapping and Physiological Significance

Table A.17 outlines the 13 anatomical keypoints selected from Google MediaPipe Pose Heavy [8, 9]. These joints constitute the core kinematic chain required for analyzing human resistance exercises. Facial landmarks (1–10) and foot extremity points (29–32) are intentionally discarded to eliminate high-frequency tracking noise.

Appendix B. Open Science Model Hub and Checkpoint Loading

All pre-trained PyTorch [30] checkpoints, scripts, and evaluation pipelines are openly accessible on Hugging Face at `https://huggingface.co/Cuong2004/gym-exercise-classification`. Checkpoints can be downloaded in Python:

Table A.17: Selected 13 anatomical joints from MediaPipe Pose and their kinematic roles.

Joint ID (v)	MediaPipe Index	Anatomical Description	Kinematic Significance in Resistance Training
0	0	Nose	Cranial orientation tracking
1	11	Left Shoulder	Glenohumeral joint; upper-body pressing/pulling rotation center
2	12	Right Shoulder	Bilateral counterpart to left shoulder
3	13	Left Elbow	Humeroradial joint; elbow flexion/extension in curls/presses
4	14	Right Elbow	Bilateral counterpart to left elbow
5	15	Left Wrist	Radiocarpal joint; barbell/dumbbell contact point trajectory
6	16	Right Wrist	Bilateral counterpart to left wrist
7	23	Left Hip	Acetabulofemoral joint; pelvic hinge pivot; midpoint with Right Hip defines coordinate origin anchor
8	24	Right Hip	Bilateral counterpart to left hip; midpoint with Left Hip defines coordinate origin anchor
9	25	Left Knee	Tibiofemoral joint; knee flexion/extension in squats/lunges
10	26	Right Knee	Bilateral counterpart to left knee
11	27	Left Ankle	Talocrural joint; ground contact anchor; ankle dorsiflexion
12	28	Right Ankle	Bilateral counterpart to left ankle

```
import torch
from huggingface_hub import hf_hub_download

# Download best Transformer Mix checkpoint
ckpt_path = hf_hub_download(
    repo_id="Cuong2004/gym-exercise-classification",
    filename="best_transformer_mix_aug.pt"
)
model = torch.load(ckpt_path, map_location="cpu")
print("Model loaded successfully!")
```